Semiadditive Thursdays

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Spring 2026

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I (2/12, Matthew Niemiro) Trivial Notions

Abstract: The greatest stories do not begin with ∞-category theory, and neither will ours—the germs of semiadditivity subsist willingly upon *ordinary* categories, ready for swab and cultivation. We will start our seminar by documenting some basic semiadditive phenomena (in particular: bilimits, enrichments, norms, and ambidexterity for negative one and zero semiadditivity) as it is expressible in ordinary language. Then we will anticipate higher extensions of the theory. Then we will create a *hostage situation* to canvas speakers.

I.1 Preamble to the seminar.

The theory of *(higher) semiadditivity* is firmly (higher) algebraic. Recent developments in chromatic and equivariant homotopy theory, trace methods, TQFTs, algebraic geometry, geometric Langlands, mirror symmetry, ... have motivated its closer study. Many folks seem familiar with some applications of semiadditivity, are interested in its other instantiations, and are not familiar with its machinations. Especially more recent work using semiadditivity requires a passing comprehension of the details to read, and this was one motivation to organize this seminar. That in mind, this seminar will take a category-first approach.

I.2 (Semi)additivity, biproducts, and commutative monoids.

Here is something you know.

Definition I.2.1. An ordinary category C is called *additive* under the following conditions.

(1) For each pair of objects $A, B \in \mathsf{C}$, the hom-set $\mathrm{Hom}_{\mathsf{C}}(A, B)$ is assigned an abelian group structure.

(2) Supposing (1), the composition of morphisms is bilinear with respect to the abelian group structure assigned to hom-sets.

(3) C admits finite biproducts (including the empty one, hence a zero object).

This is *structure*. Conditions (1) and (2) amount to an *enrichment over abelian groups*. However, some pedantic category theory dissolves the amount of choice involved. Note that such an enrichment implies that any finite product is canonically a coproduct if it exists, and vice-versa. In fact, it implies something slightly stronger: any finite (co)product is a *biproduct* $A \oplus B$ in a canonical way.[1] It likewise implies that if an initial object exists, then it is also terminal and hence a zero object. Moreover, given (1), (2), and the existence of $B \times B$ (hence the biproduct $B \oplus B$), the addition on $\mathrm{Hom}_{\mathsf{C}}(A, B)$ is uniquely determined[2] as

$$f + g : A \xrightarrow{\Delta} A \times A \xrightarrow{f \times g} B \times B \cong B \sqcup B \to B.$$

The biproduct seems to have an important role. In fact, as soon as finite biproducts exist, you can turn most of this around. If C admits finite biproducts (including a zero object), the induced operation $(f, g) \mapsto f + g$ equips C with an enrichment over commutative monoids, and this is the unique CMon-enrichment that C may possess.[3] Therefore, the definition of an additive category can be redisplayed as follows.

Definition I.2.2. An ordinary category C is called *additive* under the following conditions.

(1) C is pointed.

(2) C admits finite products and finite coproducts.

(3) Supposing (1) and (2), for each pair of objects $A, B \in \mathsf{C}$, the *identity matrix* morphism

$$\begin{pmatrix} \mathrm{id}_A & 0 \\ 0 & \mathrm{id}_B \end{pmatrix} : A \times B \to A \sqcup B$$

[1] The distinction is that biproducts are unique up to unique isomorphism while coincidental (co)products need not be, among other things. A good rabbit hole is [Yua12].

[2] This is forced by bilinearity.

[3] Two discussions of this: my older notes or [CDW24, §2].

is an isomorphism. (Thus, C admits finite biproducts.)

(4) For each pair of objects $A, B \in \mathsf{C}$, the canonical commutative monoid operation $(f, g) \mapsto f + g$ defined above on $\mathrm{Hom}_{\mathsf{C}}(A, B)$ admits negatives.

It is now clear that additivity is a *property*. (This "structure that is actually a property" thing will be a recurring theme.) Moreover, it is clear that we should inspect a weaker underlying property.

Definition I.2.3. An ordinary category possessing properties (1)-(3) is called *0-semiadditive*.

Remark I.2.4 (Prescient remark). So there's this close relationship between the *property* of semiadditivity and the *structure* of a CMon-enrichment. In particular, semiadditivity induces a canonical enrichment. The converse is false: a CMon-enrichment forces finite (co)products to be biproducts, but it does not guarantee existence. The notions coincide in the category of *pointed, finitely cocomplete* categories, however. (Or pointed, finitely complete.) We sometimes call a pointed category *(-1)-semiadditive*. We point out that pointedness can be defined just like 0-semiadditivity except with finite sets replaced by just *empty* sets, and that (-1)-semiadditivity has an analogous relationship with the "structure" of Set_*-enrichments. These are our first signs of the hierarchy of higher semiadditivity. Sam will expound upon this enrichment-property game.

(Tea and cookies break?)

I.3 Biproducts and norms.

Ordinary semiadditivity concerns the existence of finite biproducts. We can use *norms* to systematize their existence. We separate the case of empty and nonempty finite biproducts.

<u>Empty Case:</u> A terminal (resp. initial) object in C is the same thing as as limit (resp. colimit) of the empty functor $\varnothing \to \mathsf{C}$. If C admits both an initial and terminal object, denoted $\varnothing$ and $*$, then there is a unique morphism

$$\mathrm{Nm} : \varnothing \to *.$$

In this case, the invertibility of Nm is the unique condition for $\varnothing$ and $*$ to coincide. This would occur as soon as there is *any* morphism $* \to \varnothing$, which would necessarily be the inverse Nm^{-1}.

<u>Non-empty case:</u> Consider a finite family of objects $F : S \to \mathsf{C}$. We may ask for a comparison map

$$\mathrm{Nm}_F : \operatorname{colim} F \to \lim F.$$

This amounts to a family of morphisms $\mathrm{Nm}_F^{s_i,s_j} : F(s_i) \to F(s_j)$ indexed by ordered pairs $s_i, s_j \in S$, i.e. a family of choices $\mathrm{Nm}_F^{s_i,s_j} \in \mathrm{Hom}_{\mathsf{C}}(F(s_i), F(s_j))$, i.e. a matrix

	s_1	s_2	s_3	$\cdots$	s_{n-1}	s_n
s_1	$\mathrm{id}_{F(s_1)}$	?	?	$\cdots$	?	?
s_2	?	$\mathrm{id}_{F(s_2)}$	?	$\cdots$	?	?
s_3	?	?	$\mathrm{id}_{F(s_3)}$	$\cdots$	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
s_{n-1}	?	?	?	$\cdots$	$\mathrm{id}_{F(s_{n-1})}$	?
s_n	?	?	?	$\cdots$	?	$\mathrm{id}_{F(s_n)}$

Generally, there is *no* canonical Nm_F map because there are not generally distinguished elements of the mapping sets $\mathrm{Hom}_{\mathsf{C}}(F(s_i), F(s_j))$. In other words, you do not know how to fill the ? in the matrix. An easy fix is to stipulate that C is pointed, because then we can insert zero morphisms.

Remark I.3.1. The choice of a compatible family of morphisms $\varphi_{c_1,c_2} \in \mathrm{Hom}_{\mathsf{C}}(c_1, c_2)$ for each ordered pair $c_1, c_2 \in \mathsf{C}$ is equivalent to a Set_*-enrichment. Moreover, this is equivalent to "having zero morphisms." In particular, this structure is unique if it exists.

Proposition I.3.2. *If C is pointed and admits finite limits and finite colimits, then for each finite collection $F : S \to \mathsf{C}$ there is a canonical "identity matrix" morphism*

$$\mathrm{Nm}_F : \mathrm{colim}\, F \to \lim F.$$

Moreover, if Nm_F is an isomorphism, then $\lim F$ is a biproduct.

Corollary I.3.3. *For an ordinary category C, the following are equivalent.*

(1) The category C is zero semiadditive.

(2) For every finite collection $F : S \to \mathsf{C}$, the (co)limit of F exists and is a biproduct.

(3) The category C is pointed, and for every finite collection $F : S \to \mathsf{C}$, the canonical morphism $\mathrm{Nm}_F : \mathrm{colim}\, F \to \lim F$ is invertible.

We made the prior discussion to sensibly state (3). But why did we bother if we can just ask for (2)? First, let me remind you that pointedness is (-1)-semiadditivity. Next, let me pose some questions. Rather than $F : S \to \mathsf{C}$, consider a functor $F : X \to \mathsf{C}$ where now X is an anima and C is an ∞-category.

(1) What conditions guarantee that $\lim F$ or $\mathrm{colim}\, F$ exist?

(2) If they exist, what is the *best* way the limit and colimit may coincide? The distinction between biproducts and coincidental (co)products in an ordinary category is already substantial, so this should not be overlooked.

(3) What's all this good for?

The basic selling point of *higher semiadditivity* is its unreasonably effective, systematic, and broadly applicable answers to these questions. Our discussion of *ambidexterity and norms* will generalize the above analysis of 0- and (-1)-semiadditivity to handle these questions and define *higher semiadditivity*. Our discussion of *higher commutative monoids* will generalize the relation between zero semiadditivity and abelian monoids to characterize higher semiadditivity by its relation to $\mathbb{E}_\infty$ (and stronger!) structures.

I.4 Ambidexterity and biadjoints.

For a category C, semiadditivity is a property concerning every finite family $F : S \to \mathsf{C}$. Here, both F and S are variable. *Ambidexterity* conveniently isolates that part of semiadditivity concerning only S. For this, consider the map of sets

$$q : S \to *.$$

This induces a *pullback functor* $q^* : \mathsf{C} \to \mathsf{C}^S$.

Proposition I.4.1. *If C admits S-indexed (co)products, then the functors*

$$q_* := \lim_S(-) : \mathsf{C}^S \to \mathsf{C}$$
$$q_! := \mathrm{colim}_S(-) : \mathsf{C}^S \to \mathsf{C}$$

are right and left adjoints to q^* respectively. In particular, they are uniquely determined as adjoints.

Proof. The interesting counit consists of diagonals $u_{*,X} : X \to \prod_S X$ and the interesting counit consists of collapse maps $c_{!,X} : \sqcup_S X \to X$. You can now check the triangle identities. □

typos fix later

Corollary I.4.2. *For a set S, the category C admits S-indexed products iff q^* admits a right adjoint q_*, and it admits S-indexed coproducts iff q^* admits a left adjoint $q_!$.*

This gives us a "relative with respect to $S \to *$" way to talk about the existence of S-products and S-coproducts. We want to similarly understand S-bilimits.

Let's take a moment to dwell on what we are asking. We have an adjoint triple

$$q_! \dashv q^* \dashv q_*$$

and we just identified q_* and $q_!$ as computing S-indexed (co)products. Now, we want biproducts. We know that a biproduct is not just a coincidental product and coproduct, rather it needs compatible (co)product data. Similarly, we should not just ask for an equivalence of functors $q_* \cong q_!$, as this does not marry their roles as adjoints. This makes sense: if q_* (or $q_!$) is to properly serve as a simultaneous left and right adjoint, its four (co)units should be compatible. There are a few ways to go about this:

This is not implied by a functor isomorphism $q_* \cong q_!$, right?

(1) Explicit compatibilities: Suppose that C admits S-products and S-coproducts so that the adjoint triple in question exists. Then we may form

$$q^* q_* \xrightarrow{c_*} \mathrm{id}_{\mathsf{C}^S} \xrightarrow{u_!} q^* q_!.$$

Morally, if we want a good simultaneously left and right adjoint q_*, then this should compose to the identity and you should interpret this as a triangle identity. Technically, let's unwind. Note that given $F : S \to \mathsf{C}$, the functor $q^* q_!$ acts by $F \mapsto \coprod_S s_i$ and $q^* q_*$ acts by $F \mapsto \prod_S s_i$. The s_k-th component of this transformation at F is the map $i_k \circ \pi_k : \prod s_i \to \coprod s_i$. These are *precisely* the diagonal structure maps of a biproduct, which must assemble to an identity matrix to form a biproduct. In other words, we should call our adjunction *ambidextrous* if $u_! \circ c_* \cong \mathrm{id}$.

Improve this claim

I'm suspect of some of what I've written here

(2) Norms: To study biproducts, the appeal of norms is that one can bypass the fuss of compatibilities if you could form the identity matrix $\mathrm{Nm}_F : \operatorname{colim} F \to \lim F$, because its invertibility guaranteed your (co)product was a biproduct. For Nm_F to exist, we used the zero morphisms granted by pointedness. In our relative context, we may hope for a transformation

$$\mathrm{Nm}_q : q_! \to q_*$$

serving the same convenience: if we can just construct Nm_q, then its invertibility would be equivalent to the ambidexterity of our adjoint triple, i.e. the "correct" way to ask for $q_! \cong q_*$. The pause is that Nm_q may not exist: it consists of maps $\operatorname{colim}_S(-) \to \lim_S(-)$ which consist of matrices of maps, so Nm_q is only defined once you have zero morphisms for the off-diagonals, i.e. once you are (-1)-semiadditive.

(3) Wrong-way (co)units: ...

To summarize: the existence of S-(co)products amounted to the existence of adjoints $q_!$ and q_*. Then we needed pointedness to construct the norm map $\mathrm{Nm}_q : q_! \to q_*$ as the identity matrix. In this case, we say that *S is weakly C-ambidextrous*. Then if Nm_q is invertible, then C admits S-biproducts. In this case, we say that *S is C-ambidextrous*.

Proposition I.4.3. *The following are equivalent.*

(1) The set S is C-ambidextrous.

(2) The category C admits all S-indexed biproducts.

(3) The category C is pointed, and the norm $\mathrm{Nm}_q : q_! \to q_*$ associated to $q : S \to *$ is invertible.

We have delivered on our slogan: this is precisely the part of 0-semiadditivity concerning the fixed set S.

Remark I.4.4. You may replace $q : S \to *$ with an arbitrary map of finite sets $q : S \to T$. The resulting theory is as follows. The existence of adjoints is equivalent to the existence of (co)products indexed by the fibers of q; the construction of Nm_q still requires pointedness; and the invertibility of Nm_q is equivalent to the existence of biproducts indexed by the fibers of q.

Remark I.4.5. Ambidexterity, norms, and the "relative" language will become exponentially more useful in the higher setting.

II (2/19, Matthew Niemiro) Higher semiadditivity and norms

Abstract: In this session, we will make the first definition of semiadditivity using *ambidexterity and norms*. The basic idea is that semiadditivity amounts to the existence of certain bilimits, hence it is a property concerning certain families $F : X \to C$ where F and X are variable. This, in turn, may be understood inductively by analyzing the existence and invertibility of norm maps. But there's still a lot going on. The role of ambidexterity is to isolate that part of semiadditivity concerning X in a functorial and controllable manner, and this will help us run the machine. In particular, we will use ambidexterity to construct relative integration, a "higher" form of addition that will clarify the existence problem for norms. We will demonstrate this by constructing 1-finite norms for finite G-actions in the presence of ordinary semiadditivity.

II.1 Preamble to tea and cookies.

First, let's summarize parts of the previous talk which will be important today.

(1) A *biproduct* is an object serving as a product and coproduct of a family such that the canonical (co)product morphisms are compatible. If one exists, then it is unique up to unique isomorphism. This is generally distinct from an abstractly coincidental (co)product.[4] There are various equivalent definitions of biproducts, and various conditions implying their existence. If C is pointed, then biproducts admit a convenient characterization: for a set-indexed family $F : S \to \mathsf{C}$, you can use zero morphisms to define the *identity matrix* morphism

$$\mathrm{Nm}_F : \operatorname{colim} F \to \lim F,$$

and if Nm_F is invertible, then the (co)product is automatically a biproduct.

(2) The simplest definition of *ordinary semiadditivity* for a category C just asks that C admits finite biproducts. This is equivalent to demanding pointedness together with the invertibility of Nm_F for every finite-set indexed family $F : S \to \mathsf{C}$. This reformulation models the inductive definition of higher semiadditivity.

(3) Semiadditivity concerns all finite sets S and all functors $F : S \to \mathsf{C}$. *Ambidexterity* isolates the part of this property concerning only the fixed set S: just say that S is C-*ambidextrous* if C admits all S-bilimits. This may again be characterized with norms: we say that S is *weakly* C-*amibdextrous* if every $F : S \to \mathsf{C}$ admits its norm map

$$\mathrm{Nm}_F : \operatorname*{colim}_S F \to \lim_S F,$$

and if S is weakly C-ambidextrous, then it is C-ambidextrous if and only if every Nm_F is invertible. Altogether, C is 0-semiadditive if and only if every finite set S is C-ambidextrous.

(4) There is a more general and very useful way to handle ambidexterity and norms. For this, consider

$$q : S \to *.$$

The existence of S-indexed (co)limits is equivalent to the existence of adjoints $q_! \dashv q^* \dashv q_*$ to the pullback functor q^*. If C is pointed, then the norm maps for S-families assemble into a transformation $\mathrm{Nm}_q : q_! \to q_*$. Moreover, C admits S-biproducts if and only if Nm_q is invertible. We say that q is *weakly* C-*amibidextrous* if Nm_q exists, and C-*ambidextrous* if Nm_q is an isomorphism.[5] More generally, we may consider

$$q : S \to T.$$

[4] Natalie asked for an example last time. I gave the example of *the full subcategory of* Set *spanned by sets isomorphic to* $\mathbb{Z}$. Here, $A \times B$ and $A \sqcup B$ are both countably infinite, hence there are Σ_∞ isomorphisms between them as sets. However, there is no good definition of a biproduct which their (co)product can be given, and affirmingly, this category does not possess many of the good properties expected of one admitting (finite) biproducts.

[5] Notice that norms continue to serve as a convenient means for detecting biproducts: it would not be sufficient that $q_!$ and q_* are abstractly isomorphic, but if specifically $\mathrm{Nm}_q : q_! \to q_*$ exists and is invertible, then C admits S-biproducts.

Now, we say that q is *weakly* C*-ambidextrous* if $\mathrm{Nm}_q : q_! \to q_*$ exists (which requires that C is pointed and that C admits all (co)limits indexed by all the fibers of q), and we say that q is C-*ambidextrous* if Nm_q is invertible (which amounts to C being pointed and admitting all biproducts indexed by the fibers of q).

Second, let's demo the formation of *integration* using addition.

Slogan II.1.1. *The main feature of a C-ambidextrous function $f : S \to *$ is the ability to naturally integrate an S-family of morphisms $F : S \to \mathrm{Map}_{\mathsf{C}}(A, B)$ into a single morphism $\int_S F \in \mathrm{Map}_{\mathsf{C}}(A, B)$. More generally, the main feature of a C-ambidextrous function $f : S \to T$ is the ability to naturally integrate S-families of T-local systems.*

Example II.1.2 (Ordinary integration)**.** The ordinary category of abelian groups Ab admits finite bilimits, hence it is zero semiadditive. Thus, it is enriched in commutative monoids: given a finite family $f : S \to \mathrm{Hom}_{\mathsf{Ab}}(A, B)$ we may form

$$\left(\int_S f\right)(a) := \sum_{s \in S} f_s(a).$$

We can describe this categorically. First, identify f with its mate $\overline{f} : \coprod_S A \to B$. We would like to define $\int_S f$ by postcomposing this with a "diagonal into the coproduct" $A \to \coprod_S A$. We always have a morphism $\Delta : A \to \prod_S A$, and the invertibility of $\mathrm{Nm}_\varnothing : \varnothing \xrightarrow{\sim} *$ induces $\mathrm{Nm}_S : \coprod_S A \to \prod_S A$. Fortunately, the semiadditivity of Ab means that Nm_S is invertible, and we may form

$$\int_S f \quad \text{as the morphism} \quad A \xrightarrow{\Delta} \prod_S A \xrightarrow{\mathrm{Nm}_S^{-1}} \coprod_S A \xrightarrow{\overline{f}} B.$$

Now, let's frame this as a replicable and *functorial* consequence of the ambidexterity of $q : S \to *$. For this, here is a little dictionary.

Absolute notation	**Relative notation for $q : S \to *$**
$\coprod_S X$	$q_! q^* X$
$\prod_S X$	$q_* q^* X$
$\Delta_S : X \to \prod_S X$	$u_* : X \to q_* q^* X$
$f : S \to \mathrm{Map}_C(X, Y)$	$f : q^* X \to q^* Y$
$\overline{f} : \coprod_S X \to Y$	$\overline{f} : q_! q^* X \xrightarrow{q_! f} q_! q^* Y \xrightarrow{c_{!,Y}} Y$

We define *ordinary integration* as the function

$$\begin{array}{llllll}
\int_q : & \mathrm{Map}_{\mathsf{Ab}^S}(q^*A, q^*B) & \xrightarrow{q_!} & \mathrm{Map}_{\mathsf{Ab}}(q_! q^* A, q_! q^* B) & \xrightarrow{c_{!,B} \circ -} & \mathrm{Map}_{\mathsf{Ab}}(q_! q^* A, B) \xrightarrow{- \circ \mathrm{Nm}_q^{-1} \circ u_*} \mathrm{Map}_{\mathsf{Ab}}(A, B) \\
& f & \longmapsto & q_! f & \longmapsto & \overline{f} \longmapsto \overline{f} \circ \mathrm{Nm}_q^{-1} \circ u_{*,A}
\end{array}$$

What's nice is that this is obviously functorial and easy to generalize.

II.2 Higher semiadditivity, the norm problem, and ambidexterity.

A finite set is a space X such that $\pi_0(X, x_0)$ is finite and all higher $\pi_n(X, x_0)$ vanish for all $x_0 \in X$. You may be inclined to consider "higher finite sets" as follows.

Definition II.2.1. A space X is called *π-finite* if it has finite and finitely many nonzero homotopy sets. A π-finite space which is m-truncated is called *m-finite*. A space is called *(-1)-finite* if it is empty or contractible, and *(-2)-finite* if it is contractible. A map of spaces $q : X \to Y$ is called *m-finite* if its homotopy fibers are m-finite.

Remark II.2.2. There are interesting reasons to think of π-finite spaces as "higher finite sets" which we unfortunately could not get a speaker for, c.f. [**anel**].

Definition II.2.3. Just as ordinary semiadditivity asked for finite set-indexed biproducts, we say that an ∞-category C is *m-semiadditive* if every m-finite diagram $F : X \to \mathsf{C}$ admits a bilimit.

Our understanding of set-indexed bilimits readily generalizes to space-indexed bilimits. In particular,

Claim II.2.4. Let $F : X \to \mathsf{C}$ denote an m-finite diagram. Under certain hypotheses (spoiler: if X is weakly ambidextrous), there exists a natural morphism $\mathrm{Nm}_F : \operatorname{colim} F \to \lim F$ whose invertibility detects that F admits a bilimit. In particular, C is m-semiadditive if and only if it is $(m-1)$-semiadditive and the norm Nm_F associated to every m-finite diagram is invertible (i.e. every m-finite space X is ambidextrous).

What I will call "the norm problem" is the issue that Nm_F does not exist *a priori*, and when it does, its construction is inductive and a bit involved. As before, it will be convenient to take a functor calculus approach and work relative to a map $q : X \to Y$ of m-finite spaces. In this situation, we envision the norm as a transformation $\mathrm{Nm}_q : q_! \to q_*$ whose invertibility detects that *all* diagrams indexed by the fibers of q are bilimits. Toward making that precise, we should first ask:

Question II.2.5. *When do $q_!$ and q_* exist?*

We should suspect that the existence of $q_!$ and q_* depends on C admitting certain (co)limits.

Lemma II.2.6. *Given a functor between ∞-categories $q : X \to Y$, if the pullback $q^* : \mathsf{C}^Y \to \mathsf{C}^X$ admits a left adjoint $q_!$, then $q_!$ is essentially unique and is computed by the fiberwise left Kan extension*

Cite sometihng in HTT

$$q_!(F : X \to \mathsf{C}) = \left(y \mapsto \operatorname{co}\lim_{X \times_Y y} F \right).$$

Dually for the right adjoint q_* of q^*.

Corollary II.2.7. *If X and Y are anima, then the pullback $X \times_Y y$ is the homotopy fiber of q over $y \in Y$. Thus, the pullback q^* admits a left (resp. right) adjoint if and only if C admits all colimits (resp. limits) indexed by the homotopy fibers of q. In this case we say that C admits q-(co)limits.*

Example II.2.8 (Absolute case)**.** Let $q : X \to *$ denote the terminating map. An ∞-category C admits q-limits if and only if it admits all X-indexed limits $\lim(F : X \to \mathsf{C})$. Dually for q-colimits.

Now, suppose that C admits q-limits and q-colimits so that the adjoints exist.

Question II.2.9. *What does it take for $\mathrm{Nm}_q : q_! \to q_*$ to exist?*

Vague Definition II.2.10. Generalizing the ordinary case, we want to make the following language. We fix an ∞-category C and a map of spaces $q : X \to Y$.

- If Nm_q exists, then we say q is *weakly C-ambidextrous*. This map should detect q-bilimits, in the sense that if Nm_q is invertible, then C should admit all bilimits indexed by the fibers of q.
- If q is weakly ambidextrous and Nm_q is invertible, then we say q is *C-ambidextrous*.
- If $Y = *$, then we refer to the *space* X as (weakly) ambidextrous rather than the map $q : X \to *$.

Am I hallucinating, or is this actually *not* a vague definition because of some "essentially unique if it exists" fact about norms as bilimit detectors? Did I read this somewhere?

Of course, we still have to figure out when/how we can construct Nm_q. There are two equivalent ways forward, both of which will end up relying on $(m-1)$-semiadditivity. (Or at least some amount of ambidexterity.)

(1) We may construct Nm_q as a matrix whose components do not resemble an identity matrix, but rather are *integrals over certain path spaces*.

(2) We may construct the mate $\overline{\mathrm{Nm}}_q$ utilizing some special properties of the category of spaces.

Approach (1) will provide exposure therapy for integration. Approach (2) will provide exposure therapy for Beck-Chevalley conditions. We will survey both.

II.3 Norms, integration, and the definition of ambidexterity.

Let us momentarily consider the consequences of having an invertible norm map, i.e. the property of ambidexterity. We will demonstrate how this permits integration, and then we will identify how integration can be used to construct higher norm maps, starting the inductive machine.

Slogan II.3.1. *The main feature of a C-ambidextrous morphism $q : X \to *$ is that it enables the natural integration of an X-family of morphisms $F : X \to \mathrm{Map}_{\mathsf{C}}(A, B)$ into a single morphism $\int_X F \in \mathrm{Map}_{\mathsf{C}}(A, B)$. In particular, an m-semiadditive category is equipped with a natural way to integrate m-finite families*

$$\int_X : \mathrm{Map}(X, \mathrm{Map}_{\mathsf{C}}(A, B)) \to \mathrm{Map}_{\mathsf{C}}(A, B).$$

Remark II.3.2. More generally, the main feature of a C-ambidextrous map $q : X \to Y$ is the ability to naturally integrate an X-family of morphisms of Y-local systems.

Definition II.3.3. If $q : X \to Y$ is C-ambidextrous so that $\mathrm{Nm}_q : q_! \xrightarrow{\sim} q_*$ exists and is invertible, then we define *integration* relative to q as the map

$$\begin{array}{llllll} \int_q : & \mathrm{Map}_{\mathsf{C}^X}(q^*A, q^*B) & \xrightarrow{q_!} & \mathrm{Map}_{\mathsf{C}^Y}(q_!q^*A, q_!q^*B) & \xrightarrow{c_{!,B}\circ -} & \mathrm{Map}_{\mathsf{C}^Y}(q_!q^*A, B) \xrightarrow{-\circ \mathrm{Nm}_q^{-1}\circ u_*} \mathrm{Map}_{\mathsf{C}^Y}(A, B) \\ & f & \longmapsto & q_!f & \longmapsto & \overline{f} \longmapsto \overline{f} \circ \mathrm{Nm}_q^{-1} \circ u_{*,A} \end{array}$$

Now let us return to our original question and identify where integration provides an answer.

Question II.3.4. *For an m-finite space $q : X \to *$ and an ∞-category C, when does $\mathrm{Nm}_q : q_! \to q_*$ exist?*

Informally, a morphism $\mathrm{Nm}_q : q_! \to q_*$ consists of a morphism $\mathrm{Nm}_F : \operatorname{colim} F \to \lim F$ for each m-finite diagram $F : X \to \mathsf{C}$. Such a map, in turn, we understand as an "$X \times X$ matrix." More precisely, Nm_F is the data of a compatible family of morphisms

$$\mathrm{Nm}_F^{x_0,x_1} \in \mathrm{Map}_{\mathsf{C}}(F(x_0), F(x_1)) \quad \text{for every } x_0, x_1 \in X.$$

A priori, there is no canonical such map. But the diagram *does* give us a family of such maps, namely

$$F_{x_0,x_1} : \mathrm{Map}_X(x_0, x_1) \to \mathrm{Map}_{\mathsf{C}}(F(x_0), F(x_1)).$$

The big move is to observe that the path space $\mathrm{Map}_X(x_0, x_1)$ is $(m-1)$-finite. Thus, if we can integrate these $(m-1)$-finite families, then we can define Nm_q by taking $\mathrm{Nm}_F^{x_0,x_1} := \int_{\mathrm{Map}(x_0,x_1)} F_{x_0,x_1}$. This we understand to be possible if $\mathrm{Map}_X(x_0, x_1)$-families can be integrated, i.e. if $\mathrm{Map}_X(x_0, x_1)$ has an invertible associated norm, i.e. if we can call it "C-ambidextrous."

Remark II.3.5. For a general map $q : X \to Y$, we may similarly regard a morphism $\mathrm{Nm}_q : q_! \to q_*$ as a *family* of matrices varying over the base. Its component at $F : X \to \mathsf{C}$ is a morphism $\mathrm{Nm}_F : q_!F \to q_*F$ of Y-local systems, which at y specify to a morphism $\operatorname{colim}_{X_y} F \to \lim_{X_y} F$, which itself specifies an $X_y \times X_y$ matrix of morphisms $\mathrm{Nm}_{F,y}^{x_0,x_1} : F_y(x_0) \to F_y(x_1)$. We again lack a canonical choice of such a morphism, but we have a *relative path space*'s worth of choices. Informally, we should get such a morphism $\mathrm{Nm}_{F,y}^{x_0,x_1}$ for every path in X_y from x_0 to x_1, i.e. we get a *family indexed by the relative path space.*

Remark II.3.6. Each time we implicitly claim that a morphism "amounts to" the components it specifies, we depend upon an amount of naturality between the entries with respect to morphisms between their indices. This naturality exists for different reasons. Let me spitball probably incorrect justificiations:

- The assembly of $\mathrm{Nm}_{F,y}$ from $\mathrm{Nm}_{F,y}^{x_0,x_1}$ uses the universal property of (co)limits and Fubini/base-change properties of integration (which are reflections of certain Beck-Chevalley compatibilities).

- The assembly of Nm_F out of $\mathrm{Nm}_{F,y}$'s uses the functoriality of composition for integration (again, Beck-Chevalley compatibilities) and naturality of Kan extensions.
- The assembly of Nm_q out of Nm_F's uses that all the previous stuff is already functorial.

Definition II.3.7. Let C denote an ∞-category and let $q : X \to Y$ denote a map of spaces.

- For $m \geq -2$, say that q is *weakly C-ambidextrous* if q is m-truncated, C admits q-(co)limits, and
  (1) $m = -2$, in which case q is an equivalence and Nm_q is the essentially unique inverse of q^*, or
  (2) $m \geq -1$ and every $(m-1)$-finite relative path space $\mathrm{Map}_{\mathsf{C}^Y}(y_0, y_1)$ is C-ambidextrous. In this case, the norm map $\mathrm{Nm}_q : q_! \to q_*$ is the one assembled as above.
- For $m \geq -2$, say that q is *C-ambidextrous* if it is weakly C-ambidextrous and the associated Nm_q is an equivalence.

Corollary II.3.8. *A map of spaces $q : X \to Y$ is C-ambidextrous if and only if its homotopy fibers are C-ambidextrous [CSY21, Remark 2.1.2]. In particular, the definition of semiadditivity gains no generality by considering the ambidexterity of arbitrary maps $q : X \to Y$ rather than just spaces. Though, we maintain our point that this generality is useful for handling the naturality properties of norms and integration.*

Proposition II.3.9. *An ∞-category C is m-semiadditive $\iff$ every m-finite space is C-ambidextrous.*

Example II.3.10. Suppose that C is zero semiadditive. Then for every discrete space S, the map $q : S \to *$ is ambidextrous, hence every diagram $F : S \to \mathsf{C}$ admits an *invertible* norm and hence a bilimit. Moreover, for every finite group G the map $q : BG \to *$ is weakly ambidextrous, hence every object with G-action $F : BG \to \mathsf{C}$ admits a norm $\mathrm{Nm}_F : F(*)_{hG} \to F(*)^{hG}$ which may or may not be invertible.

II.4 Norms, diagonals, and Beck-Chevalley.

You may object to the amount of wishful thinking in our construction of norms (and thus, our definition of ambidexterity) using matrices. There is a more concise approach that better evidences naturality.

Given a map $q : X \to Y$, we previously imagined $\mathrm{Nm}_q : q_! \to q_*$ as an assemblage of integrals over the relative path spaces of $X \to Y$. This was useful to motivate the inductive definition of ambidexterity and its relationship to integration, but depended upon a deluded amount of hope. This was evident in our attempt to justify naturality—we could not avoid mentioning *Beck-Chevalley conditions* but did not explain them.

Let's temporarily zoom out to a simplifying level of generality.

Question II.4.1. *Suppose as given a functor*[6] $C : S^{\mathrm{op}} \to \mathsf{Cat}_\infty$ *and a commutative square in* S

$$\begin{tikzcd} A' \arrow[r, "g"] \arrow[d, "q'"'] & A \arrow[d, "q"] \\ B' \arrow[r, "f"] & B \end{tikzcd}$$

such that the functors q^* *and* $(q')^*$ *admit right adjoints* q_* *and* q'_**. What data does this situation present?*

The commutativity of the square induces an invertible 2-morphism $\alpha : g^*q^* \xrightarrow{\sim} q'^*f^*$. Our right adjoints q_* and q'_* are equivalent to their (co)unit data (u^q_*, c^q_*) and $(u^{q'}_*, c^{q'}_*)$, with which we can cook up another 2-morphism:

$$f^*q_*X \xrightarrow{u^{q'}_{f^*q_*X}} q'_*q'^*f^*q_X \xrightarrow{q'_*(\alpha^{-1})_{q_*X}} q'_*g^*q^*q_*X \xrightarrow{q'_*g^*c^q_X} q'_*g^*X.$$

This *wants* to express commutativity of the new square obtained by applying C and considering right adjoints. You should not think of this as expressing meaningful *compositional coherence* for C, rather it expresses a base-change property. Therefore, while you would expect compositional coherence, you definitely should not expect Beck-Chevalley conditions to hold in general!

Example II.4.2. The pseudofunctor $\mathsf{Set}^{/-} : \mathsf{Set}^{\mathrm{op}} \to \mathsf{Cat}$ fails to satisfy the Beck-Chevalley conditions for most non-pullback squares in Set. For an explicit example, see VIII.2.14 in my 2025 notebook. (I think this is an if and only if statement.)

Definition II.4.3. Given a commutative square $\square$ in S and a functor $C : S^{\mathrm{op}} \to \mathsf{Cat}_\infty$ as above, the *(right) Beck-Chevalley transform* associated to $\square$ is the 2-cell just defined componentwise:

$$\begin{tikzcd} A' \arrow[r, "g"] \arrow[d, "q'"'] & A \arrow[d, "q"] \\ B' \arrow[r, "f"] & B \end{tikzcd} \quad \rightsquigarrow \quad BC_* : f^*q_* \to q'_*g^*.$$

Definition II.4.4. Given C, suppose that the image q^* of every morphism $q : X \to Y$ in S admits a left adjoint (thus we might say C is *fiberwise complete*), thus a BC_* transformation. We say that C *satisfies the right Beck-Chevalley condition* if every BC_* transformation associated to a pullback square is invertible.

Definition II.4.5. Analogously, there are *left Beck-Chevalley transformations* $BC_! : f^*q_! \to q'_!g^*$.

Question II.4.6. *What did that have to do with norms?*

Proposition II.4.7 (Height 3.1.2, HL 4.3.3)**.** *Suppose as given the following pullback diagram of spaces.*

$$\begin{tikzcd} A' \arrow[r, "f"] \arrow[d, "q'"'] \arrow[dr, phantom, "\lrcorner", very near start] & A \arrow[d, "q"] \\ B' \arrow[r, "g"] & B \end{tikzcd}$$

As soon as C *admits* q*-limits (resp.* q*-colimits), the local systems functor* $\mathrm{Map}(-, \mathsf{C}) : \mathsf{S}^{\mathrm{op}} \to \mathsf{Cat}_\infty$ *satisfies the right (resp. left) Beck-Chevalley condition for this square.*

[6] If you wanted to demo this situation in ordinary category land, you would consider a *pseudofunctor* C.

For a map $q : X \to Y$ of m-finite spaces, consider the following diagram.

$$\begin{tikzcd}
X \arrow[dr, dashed, "\Delta_q"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& X \times_Y X \arrow[r, "p_2"] \arrow[d, "p_1"] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "q"] \\
& X \arrow[r, "q"] & Y
\end{tikzcd}$$

Observe that the homotopy fiber of the relative diagonal $\Delta_q : X \to X_{\times Y} X$ over $(x_0, x_1) \in X \times_Y X$ is equivalent to the space of paths identifying x_0 and x_1 within $X_{q(x_0)} = X_{q(x_1)}$. These were precisely the spaces we needed to be C-ambidextrous in order to construct Nm_q and hence define what it means for q to be *weakly* C*-ambidextrous*.

Remark II.4.8. We previously remarked that (weak) ambidexterity can be checked fiberwise. Hence, we could immediately make a (re)definition: the map $q : X \to Y$ is weakly ambidextrous if and only if Δ_q is ambidextrous, for the latter amounts to the ambidexterity of the very spaces necessary to construct Nm_q. However, this fiberwise criterion secretly relies on the Beck-Chevalley properties we are currently trying to explain.

Definition II.4.9. Let $q : X \to Y$ denote a map of spaces and suppose that C admits q-limits and q-colimits. Define the morphism

$$T : \mathrm{Map}(\mathrm{id}_{\mathsf{C}^Y}, (\Delta_q)_!(\Delta_q)^*) \to \mathrm{Map}(q_!, q_*)$$

which associates to $f : \mathrm{id}_{\mathsf{C}^Y} \to (\Delta_q)_!(\Delta_q)^*$ the composite

$$q^* q_! \xrightarrow{BC_!^{-1}} (\pi_2)_!(\pi_1)^* \xrightarrow{(\pi_2)_! f_{\pi_1^*}} (\pi_2)_!(\Delta_q)_!(\Delta_q)^*(\pi_1)^* \xrightarrow{\sim} \mathrm{id}_{\mathsf{C}^X}.$$

Definition II.4.10 (Compare Defn. 3.1.5 of [CSY22])**.** Consider a map of spaces $q : X \to Y$, an ∞-category C, and an integer $m \geq -2$.

- Say that q is *weakly* C*-ambidextrous* if C admits q-(co)limits and...
  - $m = -2$, in which case q^* is an equivalence, the essentially unique inverse is a left and right adjoint, and we define Nm_q as the identity on that inverse; or
  - $m \geq -1$ and the relative diagonal Δ_q is C-ambidextrous, in which case $\mathrm{Nm}_{\Delta_q} : (\Delta_q)_! \xrightarrow{\sim} (\Delta_q)_*$ is invertible and we define $\mathrm{Nm}_q : q_! \to q_*$ as the construction T applied to the mate of Nm_q^{-1}.
- Say that q is C*-ambidextrous* if it is weakly C-ambidextrous and the associated Nm_q is invertible.

Proposition II.4.11. *Suppose that $q : X \to Y$ is a map of m-finite spaces and that q is weakly C-ambidextrous. Then Nm_q detects q-bilimits.*

Remark II.4.12. To define weak ambidexterity, it was important for the norm Nm_{Δ_q} to be invertible, so that mating would produce a morphism $\mathrm{id} \to q_! q^*$ rather than to $q^* q_!$.

Proposition II.4.13. *This definition is equivalent to our previous one.*

III Operator's manual

Some interest was expressed in a basic semiadditivity facts sheet.

III.1 Ordinary notions

Definition III.1.1. Let C denote a pointed category and consider a set-indexed diagram $F : S \to \mathsf{C}$. We say that *F admits a biproduct Z* in the following equivalent situations. (Observe the role of zero.)

(1) The object Z is both a limit and colimit of F such that $\pi_j \circ i_k = \delta_{jk}$.

(2) The object Z is both a limit and colimit of F and the "identity matrix" morphism $\mathrm{Nm}_F : \operatorname{colim} F \to \lim F$ is invertible.

(3) The object Z is a colimit of F and the canonical maps $(\coprod_S F(s) \to F(s))_S$ exhibit $\operatorname{colim} F$ as a limit of F.

Proposition III.1.2. *When they exist, biproducts are unique up to unique isomorphism.*

Example III.1.3. The empty limit is an initial object. Dually, the empty colimit is a terminal object. The empty biproduct is a zero object.

Definition III.1.4. We say that C is *(-1)-semiadditive* if it is pointed. We say that C is *0-semiadditive* if it admits all finite bilimits.

Proposition III.1.5. *Let C denote a category and let $q : S \to *$ denote the terminal map.*

- *The right adjoint $q^* \dashv q_*$ to pullback exists $\iff$ C admits S-indexed limits.*
- *The left adjoint $q_! \dashv q^*$ to pullback exists $\iff$ C admits S-indexed colimits.*
- *If C is pointed and admits S-limits and S-colimits, then the "identity matrix" morphisms Nm_F for diagrams $F : S \to \mathsf{C}$ are natural in F and assemble to a transformation $\mathrm{Nm}_q : q_! \to q_*$. In this situation, Nm_q is invertible $\iff$ C admits S-indexed biproducts.*

Definition III.1.6. Let C denote a category. We say that a set S is *weakly C-ambidextrous* if C is pointed and admits S-indexed (co)limits, so that $\mathrm{Nm}_q : q_! \to q_*$ exists for the projection $q : S \to *$. If furthermore Nm_q is invertible so that C admits S-biproducts, then we say that S is *C-ambidextrous*.

Proposition III.1.7. *A category C is zero semiadditive $\iff$ C is pointed and every finite set S is C-ambidextrous.*

Slogan III.1.8. *The main feature of a C-ambidextrous morphism $q : S \to *$ is the ability to functorially integrate S-families of morphisms in C.*

Definition III.1.9. Suppose that S is C-ambidextrous so that Nm_F is invertible for any $F : S \to \mathsf{C}$. Given an S-family of morphisms $f : S \to \mathrm{Map}_{\mathsf{C}}(A, B)$, we define its *integral* as the morphism

$$\int_S f : A \xrightarrow{\Delta_S} \prod_S A \xrightarrow{\mathrm{Nm}_A^{-1}} \coprod_S A \xrightarrow{\overline{f}} B.$$

Remark III.1.10. All the above works for an arbitrary morphism of sets $q : S \to T$. The left adjoint $q_!$ (resp. the right adjoint q_*) exists if and only if C admits all colimits (resp. limits) indexed by the fibers of q. In this case, we say that C *admits q-colimits* (resp. q-limits). For pointed C admitting q-limits and q-colimits, fiberwise identity matrix morphisms assemble to a morphism $\mathrm{Nm}_q : q_! \to q_*$ which detects q-bilimits. Moreover, this construction is natural in q when you correctly define composition for norms [CSY21, Defn. 2.1.9 and Rmk. 2.1.10]. Integration occurs as a way to assemble S-families of T-local systems.

III.2 Higher notions

Definition III.2.1. A space X is called *π-finite* if it has finite and finitely many nonzero homotopy groups. A π-finite space which is m-truncated is called *m-finite*. A space is called *(-1)-finite* if it is empty or contractible, and *(-2)-finite* if it is contractible. A map of spaces $q : X \to Y$ is called *m-finite* if its homotopy fibers are m-finite.

Definition III.2.2. An ∞-category is called *m-semiadditive* if it admits m-finite bilimits.

Example III.2.3. Consider a map of spaces $q : X \to Y$. It is (-2)-finite $\iff$ it is an equivalence. It is (-1)-finite $\iff$ it is an inclusion of connected components.

Example III.2.4. A 1-finite space is necessarily of the form $\sqcup K(G_i, 1)$ where each G_i is a finite group.

It is practically essential to handle the theory of bilimits by working relative to maps of spaces $q : X \to Y$ and studying their associated *norms* $\mathrm{Nm}_q : q_! \to q_*$. Fortunately, this story is spiritually identical to the ordinary theory of biproducts and norms. However, the induction (which appeared nascently in the relation between pointedness and zero semiadditivity) is a bit more involved. You should mentally separate three questions:

- What does it take for the adjoints $q_!$ and q_* to exist?
- What does it take for the norm $\mathrm{Nm}_q : q_! \to q_*$ to exist?
- What are the consequences of having an *invertible* norm $\mathrm{Nm}_q : q_! \xrightarrow{\sim} q_*$? (Answer: you get integration, with which you can exhibit more norm maps.)

Proposition III.2.5. *Let $q : X \to Y$ denote a map of spaces and fix an ∞-category C. By formal Kan extension nonsense, a left adjoint $q_! \dashv q^*$ (resp. a right adjoint $q^* \dashv q_*$) necessarily computes "fiberwise (co)limits"*

$$q_!(F : X \to \mathsf{C}) = \left(y \mapsto \operatorname{co}\lim_{X \times_Y y} F \right),$$

$$q_*(F : X \to \mathsf{C}) = \left(y \mapsto \lim_{X \times_Y y} F \right).$$

Thus, the existence of q_* (resp. $q_!$) is equivalent to C admitting q-limits (resp. q-colimits).

Recall the *norm problem*: it is tricky to construct $\mathrm{Nm}_q : q_! \to q_*$. Let's unwind this for the projection $q : X \to *$. Such a morphism Nm_q should specify a morphism $\mathrm{Nm}_q(F) : q_!(F) \to q_*(F)$ for every diagram $F : X \to \mathsf{C}$. Unwinding (co)limit mapping properties, each morphism $\mathrm{Nm}_q(F)$ should in turn specify a morphism $\mathrm{Nm}_q^{a,b}(F) : F(a) \to F(b)$ for every path $a \to b$ in X. There is *a priori* no natural choice of a morphism $\mathrm{Nm}_q^{a,b}(F)$, but the diagram F induces a *family* of such choices

$$F^{a,b} : \mathrm{Map}_X(a, b) \to \mathrm{Map}_{\mathsf{C}}(F(a), F(b)).$$

The big move is to realize that if X is m-finite, then its path spaces $\mathrm{Map}_X(a, b)$ are $(m-1)$-finite. This suggests a way to handle the norm problem by inducting on truncation order: if we could *integrate* this family to a single morphism (which should be possible if $\mathrm{Map}_X(a, b)$ is ambidextrous), then we could define $\mathrm{Nm}_q^{a,b}(F) := \int_{\mathrm{Map}_X(a,b)} F^{a,b}$. Moreover, if integration is sufficiently functorial, then we can reassemble all this data into a morphism Nm_q (which should imply that q is weakly ambidextrous).

This outlines a perfectly good approach to defining ambidexterity for arbitrary maps $q : X \to Y$, with which one could then characterize higher semiadditivity in the style of Prop. III.1.7. However, there is a more concise approach that wraps everything up into a property of the relative diagonal $\Delta_q : X \to X \times_Y X$.

Definition III.2.6. Consider a map of spaces $q : X \to Y$ and an ∞-category C. Let $m \geq -2$ be an integer.

- Say that q is *m-weakly C-amibdextrous* if q is m-finite, C admits q-(co)limits, and...

- $m = -2$, in which case q^* is an equivalence and its essentially unique inverse is a biadjoint to q^*. In this case, we define $\mathrm{Nm}_q : q_! \xrightarrow{\sim} q_*$ as identity on the essentially unique inverse.
- $m \geq -1$ and the q-diagonal $\Delta_q : X \to X \times_Y X$ is $(m-1)$-C-ambidextrous. In this case, we define $\mathrm{Nm}_q : q_! \to q_*$ as the *transformation induced by* $\mathrm{Nm}_{\Delta_q}^{-1}$.

- Say that q is *m-C-ambidextrous* if it is m-weakly C-ambidextrous and its associated norm is invertible.

Proposition III.2.7. *The norms associated to a map $q : X \to Y$ are independent of m, so this parameter can be essentially ignored.*

Remark III.2.8. Recall that the *transformation induced by* $\mathrm{Nm}_{\Delta_q}^{-1}$ was defined by a mating procedure. Namely, we associated to a morphism $f : (\Delta_q)_* \to (\Delta_q)_!$ the transformation

$$q^* q_! \xrightarrow{BC_!^{-1}} (\pi_2)_!(\pi_1)^* \xrightarrow{(\pi_2)_! \overline{f}_{\pi_1^*}} (\pi_2)_!(\Delta_q)_!(\Delta_q)^*(\pi_1)^* \xrightarrow{\sim} \mathrm{id}_{\mathsf{C}^X}.$$

Then we defined $T(f) : q_! \to q_*$ as the mate of the composite. In particular, we took $\mathrm{Nm}_q := T(\mathrm{Nm}_{\Delta_q}^{-1})$.

Proposition III.2.9. *Let $q : X \to Y$ denote a map of spaces. TFAE.*

(1) The map q is C-ambidextrous.

(2) The homotopy fibers of q are C-ambidextrous.

(3) The relative path spaces of q are C-ambidextrous so that $\mathrm{Nm}_q : q_! \to q_*$ exists, and furthermore every component $\mathrm{Nm}_q(F)$ is invertible, so that C admits q-bilimits. In fact, it suffices to check those F of the form q^*G for $G \in \mathsf{C}^Y$, i.e. those which are "fiberwise constant."

Proof. todo □

Corollary III.2.10. *A category C is m-semiadditive $\iff$ every m-finite space is C-ambidextrous $\iff$ every m-finite map of spaces is C-ambidextrous.*

As before, there is the important notion of *integration*. The definition as before works quite fine, but let's phrase it categorically. This will have the added benefit of functoriality. Suppose that the terminal map $q : X \to *$ is C-ambidextrous. Here is a "dictionary."

Absolute notation	**Relative notation for $q : X \to *$**
$\mathrm{colim}_X A$	$q_! q^* A$
$\lim_X A$	$q_* q^* A$
$\Delta_S : A \to \lim_X A$	$u_{*,A} : A \to q_* q^* A$
$\nabla_S : \mathrm{colim}_X A \to A$	$c_{!,A} : q_! q^* A \to A$
$f : X \to \mathrm{Map}_C(A, B)$	$f : q^* A \to q^* B$
$\overline{f} : \mathrm{colim}_X A \to B$	$\overline{f} : q_! q^* A \xrightarrow{q_! f} q_! q^* B \xrightarrow{c_{!,Y}} B$

Definition III.2.11. Suppose that $q : X \to Y$ is C-ambidextrous. Then we define *integration along q* as the following composite.

$$\begin{aligned} \textstyle\int_q : \quad & \mathrm{Map}_{C^X}(q^*A, q^*B) \xrightarrow{q_!} \mathrm{Map}_{C^Y}(q_! q^* A, q_! q^* B) \xrightarrow{c_{!,B} \circ -} \mathrm{Map}_{C^Y}(q_! q^* A, B) \xrightarrow{-\circ \mathrm{Nm}_q^{-1} \circ u_*} \mathrm{Map}_{C^Y}(A, B) \\ & f \longmapsto q_! f \longmapsto \overline{f} \longmapsto \overline{f} \circ \mathrm{Nm}_q^{-1} \circ u_{*,A} \end{aligned}$$

Example III.2.12. Suppose that C is m-semiadditive, thus every m-finite space X is ambidextrous. We may consider a family of morphisms $f : X \to \mathrm{Map}_{\mathsf{C}}(A, B)$. The data of such f is equivalent to a morphism $q^*A \to q^*B$ where $q : X \to *$ denotes the projection. Observe that the integral $\int_q f$ unwinds into the composite

$$A \xrightarrow{\Delta_X} \lim_X A \xrightarrow{\mathrm{Nm}_q^{-1}(q^*A)} \operatorname*{colim}_X A \xrightarrow{\operatorname{colim} f} \operatorname*{colim}_X B \xrightarrow{\nabla_X} B.$$

IV (2/26, Tyler Lane) Cardinality and height

Abstract: We first review ambidexterity and higher semiadditivity via norms and integration. We then introduce p-typical semiadditivity, characterize (p-typical) semiadditivity in terms of the *cardinalities* of certain Eilenberg-Maclane spaces, and define the notion of *semiadditive height* via the cardinalities $|B^n C_p|$. We end with a brief chromatic discussion around $\mathrm{Sp}_{K(n)}$, $\mathrm{Sp}_{T(n)}$, and recollements/chromatic fracture squares.

IV.1 Preamble to tea and cookies

Ambidexterity and semiadditivity

We begin with a review of ambidexterity and semiadditivity.

Definition IV.1.1. Let A be a space and let $m \geq -1$.

- We say that A is *(-2)-finite* if A is contractible.
- We say that A is *m-finite* if A is m-truncated (i.e. $\pi_n(A, a) = 0$ for every $n > m$ and every $a \in A$) and all homotopy groups of A are finite.

We say that a map $q : A \to B$ is *m-finite* if all of its homotopy fibers are m-finite. We say that a map (or space) is *π-finite* if it is m-finite for some m.

Definition IV.1.2. Let C be an ∞-category and let $q : A \to B$ be a π-finite map of spaces.

(1) We say that q is *weakly C-ambidextrous* if either it is an equivalence, or its relative diagonal $\Delta_q : A \to A \times_B A$ is C-ambidextrous.

(2) We say that q is *C-ambidextrous* if it is weakly C-ambidextrous, C admits all q-limits and q-colimits, and the norm map $\mathrm{Nm}_q : q_! \to q_*$ is an isomorphism.

We say that a π-finite space A is (weakly) C-ambidextrous if the map $A \to \mathrm{pt}$ is.

Remark IV.1.3. Any (-2)-finite map of spaces (which is the same thing as an equivalence) is C-ambidextrous.

Definition IV.1.4. An ∞-category C is *m-semiadditive* (resp. *∞-semiadditive*) if every m-finite (resp. π-finite) map is C-ambidextrous.

Proposition IV.1.5 (Useful fact)**.** *Let $q : A \to B$ be a π-finite map. TFAE.*

(1) q is C-ambidextrous.

(2) The homotopy fibers of q are C-ambidextrous.

Corollary IV.1.6. *C is m-semiadditive if and only if every m-finite space is C-ambidextrous.*

Let's unwind what m-semiadditivity means for small m.

(-2) By definition, any ∞-category C is (-2)-semiadditive.

(-1) Any (-1)-finite space is either contractible or empty, thus (-1)-semiadditivity of C is equivalent to the C ambidexterity of $* \to *$, $\varnothing \to *$, and $\varnothing \to \varnothing$. The (weak) ambidexterity of $* \to *$ and $\varnothing \to \varnothing$ is not interesting, as these are equivalences. The weak ambidexterity of $\varnothing \to *$ amounts to the existence of initial and terminal objects $\mathrm{colim}_\varnothing = \varnothing$ and $\lim_\varnothing = *$. Then, ambidexterity is equivalent to the canonical comparison $\mathrm{Nm}_\varnothing : \varnothing \to *$ being invertible. Thus, C is (-1)-semiadditive $\iff$ C is pointed.

(0) For C to be 0-semiadditive, the empty functor must first of all be ambidextrous, hence C must be pointed. Furthermore, the weak ambidexterity of the morphism $* \sqcup * \to *$ amounts to C admitting finite (co)products. For any pair of objects $X, Y \in \mathsf{C}$, the zero maps induce an "identity matrix" morphism

$$\begin{pmatrix} \mathrm{id}_X & 0 \\ 0 & \mathrm{id}_Y \end{pmatrix} : X \amalg Y \to X \times Y$$

which is the norm in this case. Ambidexterity then amounts to this comparison being invertible. Thus, C is 0-semiadditive $\iff$ C admits finite biproducts (including a zero object).

Here is a convenient criterion for m-semiadditivity when $m > 1$. (We will later remark that in the p-typical setting, one can include $m = 1$ as well.)

Proposition IV.1.7. *Let C be an ∞-category and let $n \geq 2$. Then C is n-semiadditive if and only if*

(1) C is $(n-1)$-semiadditive, and

(2) for every prime p, the Eilenberg–MacLane space $K(\mathbb{Z}/p\mathbb{Z}, n)$ is C-ambidextrous.

Proof. We show that (1) and (2) are sufficient. Since ambidexterity is a fiberwise condition, it suffices to show that for any n-finite space X, the homotopy fibers of $X \to \tau_{\leq n-1}X$ are C-ambidextrous. We may therefore assume $X = K(A, n)$ for a finite abelian group A, and argue by induction on $|A|$. The case $A = 0$ is trivial, and $A = \mathbb{Z}/p\mathbb{Z}$ follows from (2). Otherwise choose a short exact sequence

Because this would imply those fibers of $X \to *$ are C-ambidextrous

$$0 \to A' \to A \to A'' \to 0$$

with $|A'|, |A''| < |A|$. There is a fibration $K(A', n) \to K(A, n) \to K(A'', n)$, so if $K(A', n)$ and $K(A'', n)$ are C-ambidextrous by induction, then so is $K(A, n)$. $\square$

Integration

Let C be an ∞-category and let $q : A \to B$ be a C-ambidextrous map. Then for every pair of objects $X, Y \in \mathsf{C}^B$, we can define a map

$$\int_q : \mathrm{Map}_{\mathsf{C}^A}(q^*X, q^*Y) \to \mathrm{Map}_{\mathsf{C}^B}(X, Y)$$

by letting $\int_q f$ be the composite

$$X \xrightarrow{u_*} q_*q^*X \xrightarrow{\mathrm{Nm}_q^{-1}} q_!q^*X \xrightarrow{q_!(f)} q_!q^*Y \xrightarrow{c_!} Y.$$

Remark IV.1.8. When $B = \mathrm{pt}$, we will usually write this as $\int_A$.

Definition IV.1.9. Let $\mathsf{C} \in \mathsf{Cat}_\infty$ and let $q : A \to B$ be a C-ambidextrous map. We have a natural transformation $\mathrm{id}_{\mathsf{C}^B} \xrightarrow{|q|_{\mathsf{C}}} \mathrm{id}_{\mathsf{C}^B}$ given by the composition

$$\mathrm{id}_{\mathsf{C}^B} \xrightarrow{u_*} q_*q^* \xrightarrow{\mathrm{Nm}_q^{-1}} q_!q^* \xrightarrow{c_!} \mathrm{id}_{\mathsf{C}^B}.$$

For a C-ambidextrous space A, we write $\mathrm{id}_{\mathsf{C}} \xrightarrow{|A|_{\mathsf{C}}} \mathrm{id}_{\mathsf{C}}$ and call $|A|_{\mathsf{C}}$ the *C-cardinality* of A.

For an object $X \in \mathsf{C}$, the map $X \xrightarrow{|A|_X} X$ equals $\int_A \mathrm{id}_X$. Thus we can think of $|A|_X$ as adding id_X to itself "A times", i.e. multiplying by the cardinality of A.

Example IV.1.10. Let C be a 0-semiadditive ∞-category. For every pair of objects $X, Y \in \mathsf{C}$ and every finite set A, integration gives a map

$$\prod_A \mathrm{Map}_{\mathsf{C}}(X, Y) = \mathrm{Map}_{\mathsf{C}^A}(q^*X, q^*Y) \to \mathrm{Map}_{\mathsf{C}}(X, Y).$$

In particular, taking $A = * \sqcup *$ gives a map

$$+ : \mathrm{Map}_{\mathsf{C}}(X, Y) \times \mathrm{Map}_{\mathsf{C}}(X, Y) \to \mathrm{Map}_{\mathsf{C}}(X, Y).$$

With this operation, it turns out that $\mathrm{Map}_{\mathsf{C}}(X, Y)$ admits the structure of an $\mathbb{E}_\infty$-space. If $|A| = n$, then integrating id_X over A gives

$$X \xrightarrow{\Delta_X} \prod_A X \xrightarrow{\mathrm{Nm}^{-1}} \coprod_A X \xrightarrow{\nabla_X} X,$$

i.e. the endomorphism $[n] = \mathrm{id}_X + \cdots + \mathrm{id}_X$.

Example IV.1.11. We say that an ∞-category C is *semirational* if it is 0-semiadditive and for each $n \in \mathbb{N}$, multiplication by n is invertible in C (e.g. $\mathrm{Sp}_{\mathbb{Q}}$ or $\mathsf{Vect}_{\mathbb{Q}}$). If C admits all 1-finite colimits, then C is ∞-semiadditive and for every π-finite space A,

$$|A|_{\mathsf{C}} = \sum_{a \in \pi_0(A)} \prod_{n \geq 1} |\pi_n(A, a)|^{(-1)^n}.$$

In particular,

$$|BC_p|_{\mathrm{Sp}_{\mathbb{Q}}} = \frac{1}{p} \in \mathbb{Q}_{\geq 0} \subseteq \mathrm{End}(\mathbb{S}_{\mathbb{Q}}).$$

IV.2 Height

Definition IV.2.1. Let p be a prime and $0 \leq m \leq \infty$.

(1) An ∞-category C is *p-typically m-semiadditive* if all m-finite p-spaces are C-ambidextrous.

(2) A functor $F : \mathsf{C} \to \mathsf{D}$ is *p-typically m-semiadditive* if it preserves all m-finite p-space colimits.

(3) An O-monoidal ∞-category (for an operad O) is *p-typically m-semiadditively O-monoidal* if it is p-typically m-semiadditive and compatible with m-finite p-space colimits.

We denote by $\mathsf{Cat}^{\oplus_p\text{-}m} \subset \mathsf{Cat}_\infty$ the full subcategory of p-typically m-semiadditive ∞-categories and p-typically m-semiadditive functors.

Proposition IV.2.2. *Let $0 \leq m \leq \infty$.*

(1) If $\mathsf{C} \in \mathsf{Cat}^{\oplus\text{-}0}$, then C is p-typically m-semiadditive if and only if $B^k C_p$ is C-ambidextrous for all $k = 1, \ldots, m$.

(2) For $\mathsf{C}, \mathsf{D} \in \mathsf{Cat}^{\oplus_p\text{-}m}$, a 0-semiadditive functor $F : \mathsf{C} \to \mathsf{D}$ is p-typically m-semiadditive if and only if it preserves $B^k C_p$-(co)limits for all $k = 1, \ldots, m$.

Proof. We prove the nontrivial direction of (1); (2) is similar. Assume $B^k C_p$ is C-ambidextrous for all $k = 1, \ldots, m$, and let A be an m-finite p-space. Since C is 0-semiadditive, we may reduce to the case that A is connected. The Postnikov tower of A can be refined to a tower of principal fibrations

$$A = A_r \to \cdots \to A_1 \to A_0 \to \mathrm{pt}$$

such that the fiber of each $A_i \to A_{i-1}$ is of the form $B^{k_i} C_p$ for some $1 \leq k_i \leq m$. It suffices to show each $A_i \to A_{i-1}$ is C-ambidextrous, since C-ambidextrous maps are closed under composition. Finally, since C-ambidexterity is a fiberwise condition, this follows from the C-ambidexterity of the fibers $B^{k_i} C_p$. $\square$

Definition IV.2.3. Let $\mathsf{C} \in \mathsf{Cat}^{\oplus_p\text{-}m}$. For every integer $0 \leq n \leq m$, define

$$p^{\mathsf{C}}_{(n)} := |B^n C_p|$$

as a natural endomorphism of the identity functor of C. We drop the superscript C when C is clear.

Definition IV.2.4. Let $\mathsf{C} \in \mathsf{Cat}_\infty$ and let $\alpha : \mathrm{id}_{\mathsf{C}} \to \mathrm{id}_{\mathsf{C}}$ be a natural endomorphism. An object $X \in \mathsf{C}$ is called

(1) *α-divisible* if α_X is invertible;

(2) *α-complete* if $\mathrm{Map}(Z, X) \simeq \mathrm{pt}$ for all α-divisible Z.

We write $\mathsf{C}[\alpha^{-1}]$ and $\widehat{\mathsf{C}_\alpha}$ for the full subcategories spanned by the α-divisible and α-complete objects, respectively.

Definition IV.2.5. Let $\mathsf{C} \in \mathsf{Cat}^{\oplus_p\text{-}m}$ and let $0 \le n \le m < \infty$. For every $X \in \mathsf{C}$ we define and denote the (semiadditive) *height* of X as follows:

(1) $\mathrm{ht}_{\mathsf{C}}(X) \le n$ if X is $p^{\mathsf{C}}_{(n)}$-divisible;

(2) $\mathrm{ht}_{\mathsf{C}}(X) > n$ if X is $p^{\mathsf{C}}_{(n)}$-complete;

(3) $\mathrm{ht}_{\mathsf{C}}(X) = n$ if $\mathrm{ht}_{\mathsf{C}}(X) \le n$ and $\mathrm{ht}_{\mathsf{C}}(X) > n-1$.

We extend this to $n = m = \infty$ by declaring $\mathrm{ht}_{\mathsf{C}}(X) = \infty$ iff $\mathrm{ht}_{\mathsf{C}}(X) > k$ for all $k \in \mathbb{N}$. By convention, $-1 < \mathrm{ht}_{\mathsf{C}}(X) \le \infty$ for all X, and $\mathrm{ht}_{\mathsf{C}}(X) \le -1$ (or $\mathrm{ht}_{\mathsf{C}}(X) > \infty$) iff $X = 0$. We drop the subscript C when clear.

Remark IV.2.6. The notation $\mathrm{ht}(X) \le n$ (and similarly $\mathrm{ht}(X) > n$) asserts that X satisfies a property; it does *not* mean $\mathrm{ht}(X)$ is a number being compared to n. The only object that can simultaneously have height $\le n$ and $> n$ is the zero object.

Remark IV.2.7. If C is m-semiadditive, then *a priori* we can only speak of height $> n$ for $n \le m$, since the cardinalities $|B^n C_p|$ need not make sense for $n > m$.

Example IV.2.8. Let C be a 0-semiadditive ∞-category. An object $X \in \mathsf{C}$ has height 0 iff $p = p_{(0)}$ acts invertibly on X, and has height > 0 iff it is p-complete.

Proposition IV.2.9. *Let $\mathsf{C} \in \mathsf{Cat}^{\oplus_p\text{-}m}$ and let $0 \le n_0 \le n_1 \le m$. For every $X \in \mathsf{C}$:*

(1) If $\mathrm{ht}(X) \le n_0$, then $\mathrm{ht}(X) \le n_1$.

(2) If $\mathrm{ht}(X) > n_1$, then $\mathrm{ht}(X) > n_0$.

Before proving this, we need:

Lemma IV.2.10. *Let $\mathsf{C} \in \mathsf{Cat}_\infty$ and let $q : A \to B$ be a principal C-ambidextrous map of C-ambidextrous spaces with fiber F. For $X \in \mathsf{C}$, if $|F|_X$ is invertible, then*

$$|q|_{B^*X} = |F|_{B^*X} \in \mathrm{End}(B^*X) \qquad \textit{and} \qquad |A|_X = |F|_X\,|B|_X.$$

Proof. It suffices to show: if $\mathrm{ht}(X) \le n$ for some $n \le m-1$, then $\mathrm{ht}(X) \le n+1$. Consider the principal fiber sequence

$$B^n C_p \to \mathrm{pt} \to B^{n+1} C_p.$$

All maps are C-ambidextrous by assumption. Since $\mathrm{ht}(X) \le n$, we have $|B^n C_p|_X$ invertible. By the lemma,

$$|B^{n+1} C_p|_X\,|B^n C_p|_X = |\mathrm{pt}|_X = \mathrm{id}_X,$$

hence $|B^{n+1} C_p|_X$ is invertible and $\mathrm{ht}(X) \le n+1$. The second claim follows from the first by definition. □

Remark IV.2.11. In any stable ∞-category C, a nonzero object has height 0 for at most one prime p. In particular, if C is p-local, every object has height 0 at every prime $\ell \ne p$.

Definition IV.2.12. Let $\mathsf{C} \in \mathsf{Cat}^{\oplus_p\text{-}m}$ and let $0 \le n \le m \le \infty$. Define

$$\mathsf{C}_{\le n} = \mathsf{C}[p^{-1}_{(n)}], \quad \mathsf{C}_{>n} = \widehat{\mathsf{C}_{p_{(n)}}}, \quad \mathsf{C}_n = \mathsf{C}_{\le n} \cap \mathsf{C}_{>n-1}.$$

We write $\mathrm{ht}(\mathsf{C}) \le n$, $\mathrm{ht}(\mathsf{C}) > n$, or $\mathrm{ht}(\mathsf{C}) = n$ if $\mathsf{C} = \mathsf{C}_{\le n}$, $\mathsf{C}_{>n}$, or C_n, respectively.

Lemma IV.2.13. *Let C be a p-typically m-semiadditive monoidal ∞-category. For every $0 \le n \le m$, we have $\mathrm{ht}(\mathsf{C}) \le n$ if and only if $\mathrm{ht}_{\mathsf{C}}(1) \le n$.*

Theorem IV.2.14. $\mathrm{Sp}_{K(n)}$ *and* $\mathrm{Sp}_{T(n)}$ *are ∞-semiadditive of height n.*

Theorem IV.2.15 (Redshift)**.** *Let $0 \leq n \leq m \leq \infty$ and let $\mathsf{C} \in \mathsf{Cat}_{(m+1)\text{-}fin}$. If C is p-typically m-semiadditive, then:*

(1) $\mathrm{ht}(\mathsf{C}) \leq n$ *if and only if* $\mathrm{ht}(\mathsf{C}) \leq n+1$.

(2) $\mathrm{ht}(\mathsf{C}) > n$ *if and only if* $\mathrm{ht}(\mathsf{C}) > n+1$.

Remark IV.2.16. Here ht means height as an object of the $(m+1)$-semiadditive ∞-category $\mathsf{Cat}_{(m+1)\text{-fin}}$.

Theorem IV.2.17 (Bounded height)**.** *Let C be an n-semiadditive p-local category which admits all π-finite limits and colimits. If $\mathrm{ht}(\mathsf{C}) \leq n$, then C is ∞-semiadditive.*

IV.3 Chromatic discussion

We are interested in localizations of the category $\mathrm{Sp}_{(p)}$ (and will just write Sp from now on). In particular:

(1) $\mathrm{Sp}_{K(n)}$,

(2) $\mathrm{Sp}_{T(n)}$,

(3) $L_n\mathrm{Sp} = L_{\oplus_{m=1}^n K(m)}\mathrm{Sp}$, and

(4) $L_n^f\mathrm{Sp} = L_{\oplus_{m=1}^n T(m)}\mathrm{Sp}$.

Theorem IV.3.1. $\mathrm{Sp}_{K(n)}$ *and* $\mathrm{Sp}_{T(n)}$ *are ∞-semiadditive.*

At this point, there are several proofs of this theorem which may be discussed in later talks. One might then be interested in the semiadditivity of Sp_R for other rings.

Theorem IV.3.2. *Let R be a nonzero p-local homotopy ring. Then Sp_R is 1-semiadditive if and only if there exists a (necessarily unique) integer $n \geq 0$ such that*

$$\mathrm{Sp}_{K(n)} \subseteq \mathrm{Sp}_R \subseteq \mathrm{Sp}_{T(n)}.$$

Corollary IV.3.3. Sp_R *is 1-semiadditive if and only if R is supported at a unique (finite) chromatic height.*

Thus $\mathrm{Sp}_{K(n)}$ and $\mathrm{Sp}_{T(n)}$ are the minimal and maximal 1-semiadditive localizations of Sp at rings. Since $\mathrm{Sp}_{K(n)}$ and $\mathrm{Sp}_{T(n)}$ are also ∞-semiadditive, one can ask about Sp_R:

Theorem IV.3.4. Sp_R *is 1-semiadditive if and only if* Sp_R *is ∞-semiadditive.*

This supports the following conjecture.

Conjecture IV.3.5 (Bootstrap conjecture)**.** *Every presentable stable 1-semiadditive ∞-category is ∞-semiadditive.*

By Theorem IV.3.3, the non-monochromatic localizations $L_n\mathrm{Sp}$ and $L_n^f\mathrm{Sp}$ are not 1-semiadditive. This suggests some philosophy:

(1) The semiadditivity of an ∞-category C is obstructed by nontrivial "chromatic fracture squares."

(2) There is a universal way of making an ∞-category m-semiadditive, which "dissolves the glue" in chromatic fracture squares.

Recollement

Definition IV.3.6. Let C be a stable ∞-category and $\mathsf{C}_\circ \subseteq \mathsf{C}$ a full stable subcategory. The *right orthogonal complement* $\mathsf{C}_\circ^\perp \subseteq \mathsf{C}$ is the full subcategory spanned by objects $Y \in \mathsf{C}$ such that $\mathrm{Map}(X, Y) \simeq \mathrm{pt}$ for all $X \in \mathsf{C}_\circ$.

If the inclusion $\mathsf{C}_\circ \subseteq \mathsf{C}$ admits both a left adjoint L and a right adjoint R, then C is a recollement of $\mathsf{C}_\circ$ and $\mathsf{C}_\circ^\perp$. Equivalently, C can be identified with the ∞-category of sections of the cartesian fibration of Δ^1 classified by the functor

$$L\,|_{\mathsf{C}_\circ^\perp} \colon \mathsf{C}_\circ^\perp \to \mathsf{C}_\circ,$$

which encodes the "gluing data" for constructing C from $\mathsf{C}_\circ$ and $\mathsf{C}_\circ^\perp$. When this functor is zero, there is an equivalence $\mathsf{C} \simeq \mathsf{C}_\circ \times \mathsf{C}_\circ^\perp$.

Definition IV.3.7. An inclusion of stable ∞-categories $\mathsf{C}_\circ \hookrightarrow \mathsf{C}$ that admits both a left and a right adjoint is said to *exhibit C as a recollement* of $\mathsf{C}_\circ$ and $\mathsf{C}_\circ^\perp$.

For our chromatic interests, the important source of examples of recollements arise from smashing localizations. The localizations L_n and L_n^f are smashing, and one recovers chromatic fracture squares:

Example IV.3.8.

(1) The inclusion $L_n^f\mathrm{Sp} \subseteq \mathrm{Sp}$ exhibits Sp as a recollement of $L_n^f\mathrm{Sp}$ and $\mathrm{Sp}_{F(n+1)}$ where $F(n+1)$ is a finite spectrum of type $n+1$. In particular, the inclusion $L_{n-1}^f\mathrm{Sp} \subseteq L_n^f\mathrm{Sp}$ exhibits $L_n^f\mathrm{Sp}$ as a recollement of $L_{n-1}^f\mathrm{Sp}$ and $\mathrm{Sp}_{T(n)}$.

(2) The inclusion $L_{n-1}\mathrm{Sp} \subseteq L_n\mathrm{Sp}$ exhibits $L_n\mathrm{Sp}$ as a recollement of $L_{n-1}\mathrm{Sp}$ and $\mathrm{Sp}_{K(n)}$.

Theorem IV.3.9. *Let $\mathsf{C} \in \mathsf{Cat}_{st}^{\oplus\text{-}m}$ for some $0 \le m \le \infty$. If $m < \infty$ and C is idempotent complete, then the inclusions*

$$\mathsf{C}_0, \ldots, \mathsf{C}_{m-1}, \mathsf{C}_{>m-1} \subseteq \mathsf{C}$$

determine an equivalence

$$\mathsf{C} \simeq \mathsf{C}_0 \times \cdots \times \mathsf{C}_{m-1} \times \mathsf{C}_{>m-1}.$$

When $m = \infty$ and C admits sequential limits and colimits, C is a recollement of C_∞, and one can split C as an infinite product.

Properties, structures, and modes

V (3/5, Sam Munoz-Echaniz) Spanish 101

Speaker and source

Speaker: Sam Munoz-Echaniz.
Notes: Taken live by Mattie Ji and adapted for these seminar notes.
Reference: The main reference for this talk is [**harpaz**].

Goal. Re-characterize semiadditivity in the language of spans.

Remark V.0.1. "Spans" in Spanish is written `Espán`.

V.1 Stability

Suppose that $\mathcal{C}$ has all finite colimits. Observe that in this case we have a nice way to characterize stability.

Lemma V.1.1. *$\mathcal{C}$ is stable if and only if the evaluation-at-the-sphere functor*

$$\mathrm{ev}_{\mathbb{S}}\colon \mathrm{Fun}^{ex}(\mathrm{Sp}_{\mathrm{fin}}, \mathcal{C}) \to \mathcal{C}$$

is an equivalence.

Proof idea. The proof idea is to identify $\mathrm{Fun}^{ex}(\mathrm{Sp}_{\mathrm{fin}}, \mathcal{C})$ with $\mathrm{Sp}(\mathcal{C})$. □

Now let $\mathrm{Cat}_{\mathrm{fin}}$ denote the category of categories with all finite colimits.

Lemma V.1.2. *The ∞-category of finite spectra $\mathrm{Sp}_{\mathrm{fin}}$ belongs to $\mathrm{CAlg}(\mathrm{Cat}_{\mathrm{fin}})$. In particular, Lemma V.1.1 endows a stable category $\mathcal{C}$ with the structure of a $\mathrm{Sp}_{\mathrm{fin}}$-module.*

Proof idea. Consider the diagram

$$\begin{tikzcd}
\mathrm{Sp}_{\mathrm{fin}} \times \mathcal{C} \arrow[r, "\simeq"] \arrow[d] & \mathrm{Sp}_{\mathrm{fin}} \times \mathrm{Fun}^{ex}(\mathrm{Sp}_{\mathrm{fin}}, \mathcal{C}) \arrow[dl, "(X,F)\mapsto F(X\otimes -)"] \\
\mathcal{C} &
\end{tikzcd}$$

□

Remark V.1.3. $\mathcal{S}_{\mathrm{fin}}$ (finite spaces) is the $\otimes$-unit in $\mathrm{Cat}_{\mathrm{fin}}^{\otimes}$.

Lemma V.1.4. *The functor*

$$\mathrm{Sp}_{\mathrm{fin}} \simeq \mathcal{S}_{\mathrm{fin}} \otimes \mathrm{Sp}_{\mathrm{fin}} \xrightarrow{\Sigma^\infty_+ \otimes \mathrm{Id}} \mathrm{Sp}_{\mathrm{fin}} \otimes \mathrm{Sp}_{\mathrm{fin}}$$

is an equivalence.

In particular, this implies that $\mathrm{Sp}_{\mathrm{fin}}$ is an **idempotent** object of $\mathrm{CAlg}(\mathrm{Cat}_{\mathrm{fin}})$.

Remark V.1.5. If A is idempotent, then A has a unique CAlg-structure compatible with the equivalence $A \xrightarrow{\sim} A \otimes A$.

If $M \in \mathrm{Mod}_A$, then

$$A \otimes M \xrightarrow{\simeq} M$$

is an equivalence. Indeed,

$$A \otimes M \simeq (A \otimes A) \otimes_A M \simeq A \otimes_A M \simeq M.$$

In particular, there is a fully faithful embedding

$$\mathrm{Mod}_A(\mathcal{C}) \hookrightarrow \mathcal{C}.$$

Corollary V.1.6. *This shows that*

$$\mathrm{Mod}_{\mathrm{Sp}_{\mathrm{fin}}}(\mathrm{Cat}_{\mathrm{fin}}) \hookrightarrow \mathrm{Cat}_{\mathrm{fin}}\,.$$

In other words, if $\mathcal{C}$ admits finite colimits, then being a $\mathrm{Sp}_{\mathrm{fin}}$-module is a property.

V.2 Spans

Let $\mathcal{C}$ be a category with pullbacks.

Definition V.2.1. We define $\mathrm{Span}(\mathcal{C})$ by:

(1) $\mathrm{Obj}(\mathrm{Span}(\mathcal{C})) = \mathrm{Obj}(\mathcal{C})$.

(2)
$$\mathrm{Map}_{\mathrm{Span}(\mathcal{C})}(X, Y) = \{X \leftarrow Z \rightarrow Y\} \simeq (\mathcal{C}_{/X \times Y})^{\simeq}.$$

Composition in $\mathrm{Span}(\mathcal{C})$ is given by pullback:

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ X & & Y \end{tikzcd} \quad \circ \quad \begin{tikzcd} & V \arrow[dl] \arrow[dr] & \\ Y & & W \end{tikzcd} \quad = \quad \begin{tikzcd} & & Z \times_Y V \arrow[dl] \arrow[dr] & & \\ & Z \arrow[dl] \arrow[dr] & & V \arrow[dl] \arrow[dr] & \\ X & & Y & & W \end{tikzcd}$$

Here are some key features of span categories.

Proposition V.2.2. *We have the following:*

(1) There is a functor $\mathcal{C} \to \mathrm{Span}(\mathcal{C})$ *sending* $X \mapsto X$ *and*
$$(X \to Y) \mapsto (X \xleftarrow{=} X \to Y).$$
There is also a functor $\mathcal{C}^{op} \to \mathrm{Span}(\mathcal{C})$ *given by*
$$(X \to Y) \mapsto (Y \leftarrow X \xrightarrow{=} X).$$

(2) $\mathrm{Span}(\mathcal{C})$ *is self-dual, i.e.*
$$\mathrm{Span}(\mathcal{C}) \simeq \mathrm{Span}(\mathcal{C})^{op},$$
with equivalence given by
$$(X \leftarrow Y \rightarrow Z) \mapsto (Z \leftarrow Y \rightarrow X).$$

(3) $\mathrm{Span}(\mathcal{C})$ *naturally upgrades to an* $(\infty, 2)$*-category by considering diagrams of the form*

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] \arrow[dd] & \\ X \arrow[rr, phantom, "\circlearrowright"] & & Y \\ & Z' \arrow[ul] \arrow[ur] & \end{tikzcd}$$

as 2*-morphisms.*

Spans of finite sets

Let us look at a particular example of span categories.

Example V.2.3. Let Fin denote finite sets and consider Span(Fin). An element
$$S \leftarrow M \rightarrow T$$
can be regarded as an $|S| \times |T|$ matrix with entries in $\mathbb{N}_0$, where
$$M_{s,t} = \left|\mathrm{preimage}_{(s,t)}(M \to S \times T)\right|.$$

Remark V.2.4. If you prefer not to use $\mathbb{N}_0$, you can rewrite this as an $|S| \times |T|$ matrix with entries in $\mathbb{N}$ allowing 0.

Claim V.2.5. Composition of spans corresponds to matrix multiplication.

Lemma V.2.6. Span(Fin) *is* 0*-semiadditive.*

Proof. We note that Span(Fin) is pointed with zero object $\varnothing$: for any S, there are unique maps

$$\varnothing \leftarrow \varnothing \rightarrow S \qquad \text{and} \qquad S \leftarrow \varnothing \rightarrow \varnothing.$$

We also note that coproducts exist and agree with biproducts, since Span(Fin) is self-dual. Here the coproduct $S \sqcup T$ is given by disjoint union. Indeed,

$$\begin{aligned}\mathrm{Span}(S \sqcup T, R) &= \{S \sqcup T \xleftarrow{f} M \rightarrow R\} \\ &= \{S \leftarrow M_S \rightarrow R\} \sqcup \{T \leftarrow M_T \rightarrow R\} \\ &= \mathrm{Span}(S, R) \times \mathrm{Span}(S, T),\end{aligned}$$

where $M_S = f^{-1}(S)$ and $M_T = f^{-1}(T)$. Similarly,

$$\mathrm{Span}(R, S \sqcup T) = \mathrm{Span}(R, S) \times \mathrm{Span}(R, T),$$

either by direct inspection or by self-duality. □

In fact, Span(Fin) is *the* 0-semiadditive category.

Proposition V.2.7. Span(Fin) *is the free* 0*-semiadditive category generated by* $*$.

To see this, we discuss the following proposition.

Proposition V.2.8. *Let* $\mathcal{C}$ *admit finite coproducts. Then* $\mathcal{C}$ *is* 0*-semiadditive if and only if*

$$\mathrm{ev}_*\colon \mathrm{Fun}_\oplus(\mathrm{Span}(\mathrm{Fin}), \mathcal{C}) \rightarrow \mathcal{C}$$

is an equivalence.

Proof. **Forward direction:** An object

$$F \in \mathrm{Fun}_\oplus(\mathrm{Span}(\mathrm{Fin}), \mathcal{C})$$

amounts to the following data:

- $F(\varnothing) \simeq \varnothing$, the initial object of $\mathcal{C}$.
- For a finite set S,
$$F(S) \simeq F(*)^{\oplus S}.$$
- Given a map $q\colon S \rightarrow T$ of finite sets, we have maps
$$q^*\colon F(*)^{\oplus T} \rightarrow F(*)^{\oplus S}, \qquad \text{corresponding to } T \leftarrow S \xrightarrow{=} S,$$
and
$$q_!\colon F(*)^{\oplus S} \rightarrow F(*)^{\oplus T}, \qquad \text{corresponding to } S \xleftarrow{=} S \rightarrow T.$$

In particular, for $S \in \mathrm{Fin}$, there is a map

$$F(*)^{\oplus S} \to F(\varnothing) \simeq \varnothing.$$

Thus, if $\mathcal{C}$ is semiadditive, then for each object $X \in \mathcal{C}$ one can define a functor

$$\underline{X}\colon \mathrm{Span}(\mathrm{Fin}) \to \mathcal{C}$$

by

$$\underline{X}(S) = X^{\oplus S},$$

and

$$(S \xleftarrow{p} M \xrightarrow{q} T) \mapsto \left(X^{\oplus S} \xrightarrow{p^*} X^{\oplus M} \xrightarrow{q_!} X^{\oplus T}\right).$$

Remark V.2.9. This setup can be viewed as a six-functor formalism.

One then checks that $X \mapsto \underline{X}$ is inverse to ev_*.

Converse direction: We show that

$$\mathrm{Fun}_{\oplus}(\mathrm{Span}(\mathrm{Fin}), \mathcal{C})$$

is 0-semiadditive.

First note:

- $\underline{\varnothing}$ is clearly initial.
- It is also terminal, since for
$$F \in \mathrm{Fun}_{\oplus}(\mathrm{Span}(\mathrm{Fin}), \mathcal{C}),$$
the unique map $F(*) \to \varnothing$ induces a map $F \to \underline{\varnothing}$.
- Hence this functor category is pointed.

Now we wish to show that products and coproducts agree. Since

$$\mathrm{Fun}_{\oplus}(\mathrm{Span}(\mathrm{Fin}), \mathcal{C}) \simeq \mathcal{C},$$

we may identify such a functor F with the object $F(*)$.

Take

$$F, G \in \mathrm{Fun}_{\oplus}(\mathrm{Span}(\mathrm{Fin}), \mathcal{C}).$$

We want to show that $F \oplus G$ is a product. There are projection maps

$$p_F\colon F \oplus G \to F \oplus \underline{\varnothing} = F, \qquad p_G\colon F \oplus G \to \underline{\varnothing} \oplus G = G.$$

This yields the diagram

$$\begin{tikzcd}
H \arrow[rr] \arrow[dr, dashed] \arrow[ddr] & & G \\
& F \oplus G \arrow[ur] \arrow[d] & \\
& F &
\end{tikzcd}
\qquad
\begin{array}{ccc}
H \longrightarrow F & \leftrightarrow & H(*) \xrightarrow{f} F(*) \\
H \longrightarrow G & \leftrightarrow & H(*) \xrightarrow{g} G(*)
\end{array}$$

By restricting H to the span

$$\underline{1} \leftarrow \underline{2} \to \underline{2},$$

this gives a map

$$H \to H \oplus H \xrightarrow{f \oplus g} F \oplus G \qquad (\dagger).$$

One checks that the dashed map in the diagram is precisely $(\dagger)$. This shows that $F \oplus G$ has the desired universal property of the product. $\square$

Spans of π-finite spaces

Notation.

- $\mathcal{S}_m \subset \mathcal{S}$ denotes the full subcategory on m-finite spaces. Here an m-finite space is an m-truncated space all of whose homotopy groups are finite.
- $\mathcal{K}_m = \pi_0(\mathcal{S}_m^{\simeq})$ is the collection of homotopy types of m-finite spaces.
- For $-2 \leq n \leq m$, $\mathcal{S}_{m,n} \subset \mathcal{S}_m$ denotes the wide subcategory on n-truncated maps.
- We write
$$\mathcal{S}_m^n = \mathrm{Span}(\mathcal{S}_m, \mathcal{S}_{m,n}) \subset \mathrm{Span}(\mathcal{S}_m)$$
for the span category whose morphisms are spans
$$X \underbrace{\leftarrow}_{n\text{-truncated}} Z \to Y.$$
In particular,
$$\mathcal{S}_m^m = \mathrm{Span}(\mathcal{S}_m).$$
- We write $\mathrm{Cat}_{\mathcal{K}_m}$ for the category of categories admitting $\mathcal{K}_m$-indexed colimits.

Example V.2.10. Being a (-2)-truncated map of spaces is the same as being an equivalence, so
$$\mathcal{S}_{m,-2} \simeq \mathcal{S}_m^{\simeq}.$$

The following is a fact whose proof we omit.

Proposition V.2.11. $\mathrm{Cat}_{\mathcal{K}_m}^{\otimes}$ *is symmetric monoidal, with tensor product* $\otimes_{\mathcal{K}_m}$ *characterized by*
$$\mathrm{Fun}_{\mathcal{K}_m}(\mathcal{C} \otimes_{\mathcal{K}_m} \mathcal{D}, \mathcal{E}) \simeq \mathrm{Fun}_{\mathcal{K}_m,\mathcal{K}_m}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}).$$

Proposition V.2.12. $\mathcal{S}_m^n$ *is an object of* $\mathrm{CAlg}(\mathrm{Cat}_{\mathcal{K}_m}^{\otimes})$ *under pointwise product, i.e.*
$$(X \leftarrow W \to Y) \cdot (X' \leftarrow W' \to Y') = (X \times X' \leftarrow W \times W' \to Y \times Y').$$
This is well-defined because the wrong-way maps are n-truncated.

Words on the proof. This follows from the case $n = -2$. In that case,
$$\mathcal{S}_m^{-2} = \mathcal{S}_m^{\simeq}$$
is already symmetric monoidal. One then considers the product
$$\mathcal{S}_m^n \times \mathcal{S}_m^n \to \mathcal{S}_m^n$$
and observes that every $\mathcal{K}_n$-diagram in $\mathcal{S}_m^n$ comes from $\mathcal{S}_m$. □

V.3 Higher semiadditivity and spans

Semiadditivity in terms of spans

Theorem V.3.1. *If*
$$\mathcal{D} \in \mathrm{Mod}_{\mathcal{S}_m^m}(\mathrm{Cat}_{\mathcal{K}_m}),$$
then $\mathcal{D}$ is m-semiadditive.

To prove this theorem, we set up the following notation:

- If $\mathcal{D} \in \mathrm{Mod}_{\mathcal{S}^n_m}$, we have an action map
$$\mathcal{S}^n_m \times \mathcal{D} \to \mathcal{D}$$
which is $\mathcal{K}_m$-bilinear, i.e. preserves $\mathcal{K}_m$-colimits in each variable. For $x \in \mathcal{S}^n_m$, we obtain a map
$$[x] \colon \mathcal{D} \to \mathcal{D}, \qquad d \mapsto x \otimes d.$$

- For $x \in \mathcal{S}^n_m$, we have spans
$$\mathrm{tr}_x \colon X \times X \xleftarrow{\Delta} X \to *, \qquad \Delta_x \colon * \leftarrow X \xrightarrow{\Delta} X \times X.$$
These induce natural transformations
$$[\mathrm{tr}_x] \colon [x] \circ [x] \simeq [x \times x] \to [*] \simeq \mathrm{id}_{\mathcal{D}},$$
and
$$[\Delta_x] \colon \mathrm{id}_{\mathcal{D}} \to [x] \circ [x].$$

Proposition V.3.1. *Suppose $\mathcal{D} \in \mathrm{Cat}_{\mathcal{K}_m}$ such that:*

- *$\mathcal{D}$ is $(m-1)$-semiadditive.*
- *$\mathcal{D} \in \mathrm{Mod}_{\mathcal{S}^{m-1}_m}(\mathrm{Cat}_{\mathcal{K}_m})$.*

Then $\mathcal{D}$ is m-semiadditive if and only if the natural transformation $[\mathrm{tr}_x]$ exhibits multiplication by x, namely $[x]$, as self-adjoint for all x.

This proposition takes about 10–15 pages of the paper to prove, so we omit the proof.

Example V.3.2. As a sanity check, when $m = 0$ we are considering the identity matrix
$$I_n \colon d^{\oplus n} \to d^{\times n}.$$
The condition of being "self-adjoint" should be thought of as a kind of nondegeneracy condition.

Proof of Theorem V.3.1. We show that $\mathcal{D}$ is n-semiadditive for all n such that $-2 \le n \le m$. Suppose by induction that $\mathcal{D}$ is n-semiadditive for $n < m$. The proposition above reduces the claim to showing that for every $x \in \mathcal{S}^n_{n+1}$, the map
$$[\mathrm{tr}_x] \colon [x] \circ [x] \to \mathrm{Id}$$
is the counit of an adjunction with unit
$$[\Delta_x] \colon \mathrm{Id} \to [x] \circ [x].$$
These maps make sense because $\mathcal{D}$ is an $\mathrm{Mod}_{\mathcal{S}^m_m}(\mathrm{Cat}_{\mathcal{K}_m})$.

We verify one triangle identity. Consider
$$\begin{tikzcd}
{[x]} \arrow[r, "{[x]([\Delta_x])}"] \arrow[dr, "="'] & {[x] \circ [x] \circ [x]} \arrow[d, "{[\mathrm{tr}_x]_{[x]}}"] \\
& {[x]}
\end{tikzcd}$$

Observe that in $\mathrm{Span}(\mathcal{S}^m_{n+1})$ we have a diagram
$$\begin{tikzcd}
& & X \arrow[dl, "\Delta"'] \arrow[dr, "\Delta"] & & \\
& X \times X \arrow[dl, "pr_1"'] \arrow[dr, "1 \times \Delta"] & \text{pullback} & X \times X \arrow[dl, "\Delta \times 1"] \arrow[dr, "pr_2"] & \\
X & & X \times X \times X & & X
\end{tikzcd}$$
such that $pr_1 \circ \Delta$ and $pr_2 \circ \Delta$ are identities. This verifies the desired triangle identity. The other triangle identity is similar. $\square$

Corollary V.3.3. *For all $-2 \leq n \leq m$, the object*

$$\mathcal{S}_m^n \in \mathrm{CAlg}(\mathrm{Cat}_{\mathcal{K}_m})$$

is n-semiadditive.

Proof. $\mathcal{S}_m^n$ is clearly an $\mathcal{S}_m^n$-module, so the theorem applies. □

Corollary V.3.4. *This recovers the earlier statement that* Span(Fin) *is* 0*-semiadditive.*

Modules in terms of semiadditivity

Theorem V.3.2 (Universal property of m-spans)**.** *For $-2 \leq n \leq m$, suppose $\mathcal{D} \in \mathrm{Cat}_{\mathcal{K}_m}$ is n-semiadditive. Then the following is an equivalence:*

$$\mathrm{ev}_*\colon \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^n, \mathcal{D}) \xrightarrow{\simeq} \mathcal{D}.$$

Remark V.3.5. The converse is also true. This equivalence shows that $\mathcal{D}$ becomes a module over $\mathcal{S}_m^n$, via

$$\mathcal{S}_m^n \times \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^n, \mathcal{D}) \to \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^n, \mathcal{D}),$$

$$(X, F) \mapsto F(X \times -).$$

Since this is a module category, it is therefore n-semiadditive.

We omit the proof of Theorem V.3.2.

V.4 Consequences

The category of m-semiadditive categories

Our goal is now to describe the category Add_m as the full subcategory of $\mathrm{Cat}_{\mathcal{K}_m}$ on m-semiadditive categories.

Consider the forgetful functor

$$\mathcal{U}\colon \mathrm{Mod}_{\mathcal{S}_m^m}(\mathrm{Cat}_{\mathcal{K}_m}) \to \mathrm{Cat}_{\mathcal{K}_m}.$$

Observation.

$$\mathrm{Im}(\mathcal{U}) = \mathrm{Add}_m.$$

In particular, $\mathcal{U}$ sits in an adjunction

$$\mathcal{U}\colon \mathrm{Mod}_{\mathcal{S}_m^m}(\mathrm{Cat}_{\mathcal{K}_m}) \rightleftharpoons \mathrm{Cat}_{\mathcal{K}_m} : \mathcal{S}_m^m \otimes_{\mathcal{K}_m} (-).$$

This gives the following proposition.

Proposition V.4.1. *(1) For $\mathcal{C} \in \mathrm{Cat}_{\mathcal{K}_m}$, the category $\mathcal{C}$ is m-semiadditive if and only if the unit*

$$\mathcal{C} \simeq \mathcal{S}_m \otimes_{\mathcal{K}_m} \mathcal{C} \to \mathcal{S}_m^m \otimes_{\mathcal{K}_m} \mathcal{C}$$

is an equivalence.

(2) If $\mathcal{C} \in \mathrm{Mod}_{\mathcal{S}_m^m}$, then the counit map

$$\nu\colon \mathcal{S}_m^m \otimes_{\mathcal{K}_m} \mathcal{C} \to \mathcal{C}$$

is an equivalence. In particular, $\mathcal{U}$ is fully faithful.

Corollary V.4.2.

$$\mathrm{Mod}_{\mathcal{S}_m^m}(\mathrm{Cat}_{\mathcal{K}_m}) \simeq \mathrm{Add}_m.$$

Proof of Proposition V.4.1. For part (1), the converse follows because the displayed condition equips $\mathcal{C}$ with the structure of an $\mathcal{S}_m^m$-module, hence $\mathcal{C}$ is m-semiadditive by Theorem V.3.1.

For the forward implication, since $\mathcal{S}_m^m \otimes \mathcal{C}$ is m-semiadditive, it suffices to show that for all $\mathcal{D} \in \mathrm{Add}_m$, the diagonal map is an equivalence:

$$\begin{tikzcd}
\mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^m \otimes \mathcal{C}, \mathcal{D}) \arrow[r, "\cong"] \arrow[d, "\cong"'] & \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{C}, \mathcal{D}) \\
\mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^m, \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{C}, \mathcal{D})) \arrow[ur, "\mathrm{ev}_*"'] &
\end{tikzcd}$$

This is an equivalence because $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is an $\mathcal{S}_m^m$-module, hence m-semiadditive.

For part (2), factor the identity as

$$\mathrm{id}_{\mathcal{C}} \colon \mathcal{C} \xrightarrow{u} \mathcal{S}_m^m \otimes \mathcal{C} \xrightarrow{\nu} \mathcal{C}.$$

It suffices to show that u is an equivalence, which follows from part (1). □

Let us record some further corollaries.

Corollary V.4.3. *(1) The forgetful functor*

$$\mathcal{U} \colon \mathrm{Mod}_{\mathcal{S}_m^m}(\mathrm{Cat}_{\mathcal{K}_m}) \to \mathrm{Cat}_{\mathcal{K}_m}$$

has right adjoint

$$\mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^m, -).$$

(2) The object

$$\mathcal{S}_m^m \in \mathrm{CAlg}(\mathrm{Cat}_{\mathcal{K}_m})$$

is an idempotent algebra. That is, the multiplication map

$$\times \colon \mathcal{S}_m^m \times \mathcal{S}_m^m \xrightarrow{\simeq} \mathcal{S}_m^m$$

is an equivalence.

Proof. Part (2) follows directly from Proposition V.4.1. Part (1) follows from the fact that

$$\mathrm{Fun}_{\mathcal{K}_m}\big(\mathcal{C}, \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^m, \mathcal{D})\big) \simeq \mathrm{Fun}_{\mathcal{K}_m}\big(\mathcal{S}_m^m \otimes_{\mathcal{K}_m} \mathcal{C}, \mathcal{D}\big).$$

□

Higher commutative monoids

Let us begin with an observation.

Remark V.4.4. $\mathcal{D}$ is m-semiadditive if and only if $\mathcal{D}^{op}$ is m-semiadditive. Indeed, $\mathcal{S}_m^m$-modules are the same as $(\mathcal{S}_m^m)^{op}$-modules.

Thus we have a diagram

$$\begin{tikzcd}
\mathrm{Add}_m \arrow[r] \arrow[d, "(-)^{op}, \sim"] & \mathrm{Cat}_{\mathcal{K}_m} \arrow[d, "(-)^{op}, \simeq"'] \\
\mathrm{Add}_m \arrow[r] & \mathrm{Cat}^{\mathcal{K}_m}
\end{tikzcd} \qquad \text{categories with } \mathcal{K}_m\text{-limits}$$

So all previous results for m-semiadditivity also work for categories in $\mathrm{Cat}^{\mathcal{K}_m}$.

Example V.4.5. If $\mathcal{C} \in \mathrm{Cat}^{\mathcal{K}_m}$, then

$$\mathrm{Fun}^{\mathcal{K}_m}(\mathcal{S}_m^m, \mathcal{C})$$

is the universal m-semiadditive category equipped with a limit-preserving functor to $\mathcal{C}$.

Proof. We have

$$\mathrm{Fun}^{\mathcal{K}_m}(\mathcal{S}_m^m, \mathcal{C}) \simeq \mathrm{Fun}^{\mathcal{K}_m}((\mathcal{S}_m^m)^{op}, \mathcal{C}) \simeq \mathrm{Fun}_{\mathcal{K}_m}(\mathcal{S}_m^m, \mathcal{C}^{op})^{op}.$$

□

Definition V.4.6. Let $\mathcal{D} \in \mathrm{Cat}^{\mathcal{K}_m}$. An *$m$-commutative monoid* in $\mathcal{D}$ is a functor

$$F \in \mathrm{Fun}(\mathcal{S}_m^{m-1}, \mathcal{D})$$

such that the diagram

$$\{F(x \xleftarrow{i_x} \{x\} \to *) \colon F(*) \to F(x)\}_{x \in X}$$

exhibits

$$F(X) \xrightarrow{\simeq} \lim_{x \in X} F(*).$$

We write $\mathrm{CMon}_m(\mathcal{D})$ for the full subcategory of $\mathrm{Fun}(\mathcal{S}_m^{m-1}, \mathcal{D})$ spanned by the m-commutative monoids.

Example V.4.7. We have:

(1) When $m = -1$, one has

$$\mathrm{CMon}_{-1}(\mathcal{D}) \simeq \mathcal{D}_*,$$

the category of pointed objects of $\mathcal{D}$.

(2) When $m = 0$, recall that $\mathcal{S}_0^{-1} \subset \mathrm{Span}(\mathrm{Fin})$ is the full subcategory on spans

$$S \hookleftarrow M \to T,$$

where the wrong-way map is injective.
This is equivalent to the data of a based map

$$\alpha \colon S_+ = S \sqcup \{*\} \to T_+.$$

Here $\alpha(*) = *$, while $\alpha(s) = q(m)$ if $m = p(s)$, and $\alpha(s) = *$ otherwise.
Thus $\mathrm{CMon}_0(\mathcal{D})$ may be identified with the full subcategory of

$$\mathrm{Fun}(\mathrm{Fin}_*, \mathcal{D})$$

consisting of those F for which

$$F(n) \simeq F(*)^{\times n}.$$

In other words,

$$\mathrm{CMon}_0(\mathcal{D}) \simeq \mathrm{CMon}(\mathcal{D}) = \mathrm{Alg}_{\mathbb{E}_\infty}(\mathcal{D}^\times),$$

the usual commutative monoids in $\mathcal{D}$.

(3) Let $\mathcal{D} = \mathcal{S}$ be spaces, and let

$$F \in \mathrm{CMon}_m(\mathcal{D}).$$

We think of $M = F(*)$ as the underlying object of the commutative monoid F. For $X \in \mathcal{S}_m^{m-1}$, the limit condition says

$$F(X) \simeq \lim_X F(*) \simeq \mathrm{Map}(X, M).$$

There are two important kinds of spans in $\mathcal{S}_m^{m-1}$.

First, a span of the form

$$Y \xleftarrow{f} X \xrightarrow{=} X$$

induces

$$f^*\colon F(Y) = \mathrm{Map}(Y, M) \to F(X) = \mathrm{Map}(X, M),$$

namely precomposition by f.
Second, a span of the form

$$X \xleftarrow{=} X \to *$$

induces an "integration" map

$$\int_X : \mathrm{Map}(X, M) \to M.$$

There are also spans of the form

$$X \xleftarrow{=} X \to Y,$$

which let the target Y control how this integration behaves, though we will not pursue that here.

We end the lecture with the following theorem.

Theorem V.4.1. *For $m \geq -1$, let $\mathcal{D} \in \mathrm{Cat}^{\mathcal{K}_m}$. Then the natural functor*

$$\mathrm{Fun}^{\mathcal{K}_m}(\mathcal{S}_m^m, \mathcal{D}) \to \mathrm{Fun}(\mathcal{S}_m^{m-1}, \mathcal{D})$$

lands in $\mathrm{CMon}_m(\mathcal{D})$ and is an equivalence. In particular, if $\mathcal{D}$ is m-semiadditive, then

$$\mathcal{D} \simeq \mathrm{CMon}_m(\mathcal{D}).$$

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